Document Control

Title			
Acute Pain Guidelines			
Author		**Author's job title** Consultant In Anaesthesia and Pain Management, Acute Pain Lead	
Directorate Planned Care		**Department** Anaesthetics	
Version	**Date Issued**	**Status**	**Comment / Changes / Approval**
0.1	Nov 2008	Draft	Reviewed and re-formatted into corporate template.
0.1	Nov 2008	Draft	Sent out for consultation with stakeholders. Approved without amends.
1.0	Nov 2008	Final	Final approval given by the Drugs and Therapeutics Group and the Director of Pharmaceutical Services
1.1	Nov 2009	Revision	Revision made due to introduction of new analgesic techniques and discontinuation of epidural pre-filled bags (Bupivacaine 0.1%). Sent out for consultation with stakeholders. Approved by stakeholders without amends.
1.2	Jan 2010	Revision	Minor amendments by Corporate Affairs to formatting and document control report.
1.3	Aug 2010	Revision	Corrected document control report. Amends to formatting for document map viewing. Updated appendices A & B.
1.4	Aug 2010	Revision	Revision to section 10 (epidural analgesia), guidelines of monitoring of motor function added (Appendix D).
1.5	Sep 2011	Revision	Updated best practice guidelines for Entonox in accordance with BOC medical gases recommendations.
1.6	Sep 2011	Revision	Added new prescription Continuous Nerve and Wound infusion prescription. Approved by the Lead Consultant Anaesthetist on 6th September.
1.7	Sep 2011	Revision	Minor amendments by Corporate Governance to document control report, headers and footers, update Trust title, formatting for document map navigation.
2.0	Jan 2013	Final	Approved by Drugs and Therapeutics Committee on 17th January.
2.1	Jan 2013	Revision	Minor amendments by Corporate Governance to version control, formatting for document map navigation and semi-automatic table of contents. Updated Trust logo on forms (Appendices A, B, C & D). Corrected role title.
3.0	Jan 2017	Final	Significant revision. Approved by Drugs and Therapeutics Committee on 19th January 2017.
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Superseded Documents N/A		
Issue Date Jan 2017	**Review Date** Jan 2020	**Review Cycle** Three years
Consulted with the following stakeholders: (list all) • Consultant Anaesthetists • Paediatric Doctors and Nurses • Lead Surgical Pharmacist and Clinical Pharmacy Manager • Drugs and Therapeutics Committee • Recovery Staff • Patient Documentation Group		
Approval and Review Process • Drugs and Therapeutics Committee		
Local Archive Reference G:\Corporate Governance\Published Policies\Anaesthetics **Local Path** Policies\Anaestheticsfolder **Filename** Acute Pain Guidelines v3.0 19Jan17.doc		
Policy categories for Trust's internal website (Bob) Anaesthetics / Acute Pain	**Tags for Trust's internal website (Bob)** PCA / Epidural / Ketamine Infusion / Opioids / Nerve Catheter	

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1. Purpose of Acute Pain Guidelines

The following general principles can be applied in order to improve pain management and reduce potential errors in drug prescribing and administration. They should be used in conjunction with the section on General Analgesic Prescribing (see Section 5).

Assess pain regularly and fully, using the Trust-wide pain assessment tool whenever possible.

Develop a consistent, standard approach to analgesic prescribing using a few drugs with which dosing and side effects are familiar to the prescriber.

Consider the need for opioids carefully. In most patients it is straight forward and related to pain intensity. In opioid dependent patients and patients with (benign) persistent pain, the decision is more complicated and it is wise to involve the Pain Team as soon as possible.

Use the Acute Pain Team (Bleep 279):

- Band 6 level Clinical Nurse Specialists who will advise within these guidelines.
- Band 7 Clinical Nurse Specialists (qualified non-medical prescriber) who may give independent prescribing advice and provide leadership to the band 6 CNS.
- Consultant Anaesthetist specialising in Pain Medicine (pain consultant).

Out of hours, the on-call anaesthetic team provide support for pain issues and are contacted on Bleep 504.

Consider involving the Palliative Care Team for those patients who suffer pain with advanced terminal illness (Bleep 401).

This guideline applies to all healthcare professionals and deviations from it need to be clinically justified. All drug doses recommended are in accordance with the BNF or agreed simplifications. This guideline is not a substitute for the BNF, which should be consulted regarding precautions for all drugs covered by this guideline. In case of confliction between the BNF and this guideline consult the Acute Pain Team. Default to BNF advice if doubt persists.

2. Definitions

- AKI — Acute Kidney Injury
- BNF …….. — British National Formulary
- CNS ……. — Clinical Nurse Specialist
- g ..... — Gram(s)
- GP …… — General Practitioner
- IM ..... — Intra-muscular
- IR ..…… — Immediate Release preparation
- IV..... — Intravenous
- kg ..... — Kilograms

- mcg…… Micrograms
- MC&S..... Microscopy, Culture and Sensitivity
- mg ..... Milligrams
- ml ..... Millilitres
- NSAID ..... Non-Steroidal Anti-Inflammatory Drug
- NCA ……. Nurse Controlled Analgesia
- PCA ..... Patient Controlled Analgesia
- PO.....…….. Oral dose
- PR ..... Rectal dose
- SC ..... Subcutaneous
- SR …… Sustained Release preparation
- TENS ..... Transcutaneous Electrical Nerve Stimulation
- VAS..... Visual Analogue Score

3. Responsibilities

3.1. The Associate Director for Planned Care is responsible for the guidelines but will delegate functional responsibility to the Lead Consultant for Acute Pain. The guidelines are approved by the Drugs and Therapeutics Committee. The guidelines are to be implemented by Medical and Nursing staff involved in the assessment and treatment of acute pain.

3.2. Role of Lead Consultant for Acute Pain

The Lead Consultant for Acute Pain is responsible for:

- Ensuring that guidelines are in line with modern practice and that dosing regimens are appropriate.
- Ensuring that safety mechanisms are in place for advanced pain techniques.

3.3. Role of Drugs and Therapeutics Group

The Drugs and Therapeutics Group is responsible for:

- Acting as a referee to check that recommended dosing regimens are safe.
- Ensuring that safety mechanisms are in place for advanced pain techniques.

4. Introduction

This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines to enable safe and effective management of acute pain across the whole organisation. Many patients present with pain whilst in hospital. Acute pain may be related to a patient's condition, surgery or trauma.

The commonly accepted definition of pain is that of the International Association of the Study of Pain (IASP) who describe it as 'an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage.'

Acute illness, surgery and trauma are strongly associated with tissue damage and pain.

Pain is generally considered to be nociceptive when it arises from tissue damage and subsequent activation of pain nociceptors. It is considered to be neuropathic if the pain arises from neuronal damage, either from peripheral nerves or within the central nervous system.

In major surgery and trauma pain may involve a mixture of neuronal and non-neuronal tissue damage and hence both nociceptive and neuropathic pain should be considered, especially if wounds cut across nerve fibres. Otherwise, unless there is identified nerve injury, nociceptive pain usually predominates.

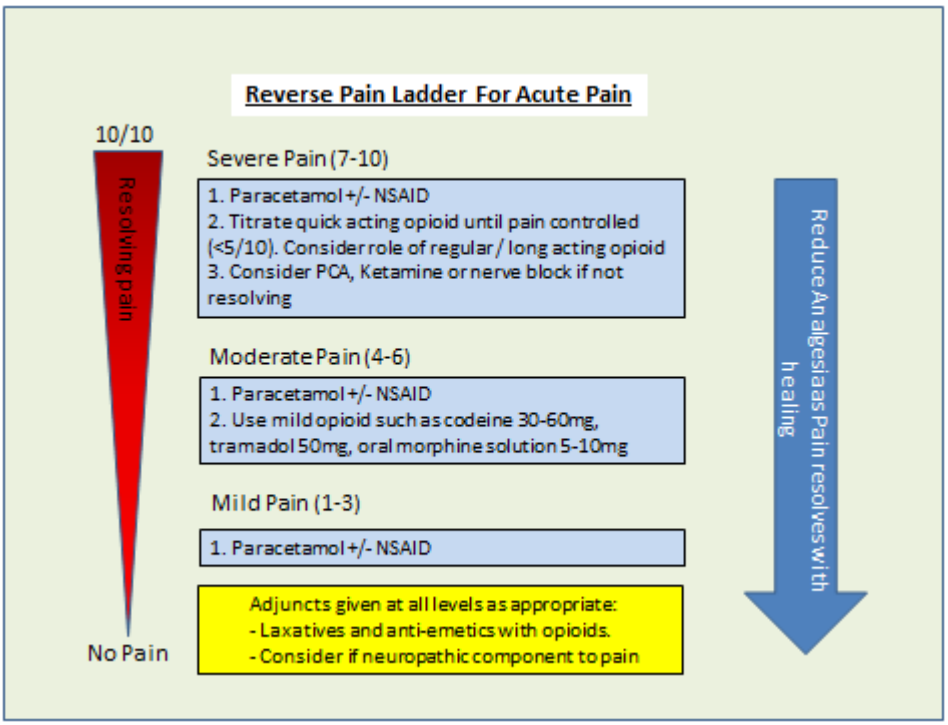
The treatment of pain is dependent on its severity and type and therefore must be assessed appropriately to be managed correctly. As pain is a dynamic experience and will vary in intensity, this must be done frequently and recorded, along with the patient's vital signs.

Episodes of moderate to severe pain (>5 /10) require treatment promptly. Evidence that this has occurred and been followed-up must be documented.

The basis of acute pain therapy is the 'reverse analgesic ladder'. This is an adaptation of the well-known World Health Organisation (WHO) pain ladder which was designed specifically for steadily increasing cancer pain.

The reverse ladder assumes pain to be more severe immediately following surgery, trauma or acute medical illness and then reduces with time and healing, allowing a reduction in analgesia.

This needs to be used in conjunction with neuropathic pain medication (where required) and regional anaesthesia techniques where applicable. The ladder can be summarised as follows:



5. Analgesic Prescribing – General Principles

5.1. Assessment of Pain, Analgesia, Sedation, Nausea and Vomiting

Before starting or altering pain relief, an assessment of the degree of pain should be made using a validated rating scale. The most common scoring system used in NDDH is a 0-10 scale:

PAIN SCORE	DESCRIPTION AT REST	DESCRIPTION ON MOVEMENT
10	Worst pain imaginable	Worst pain imaginable
7-9	Severe	Severe and Persisting
4-6	Moderate	May be briefly severe
1-3	Mild	May be briefly moderate
0	Nil	May be mild

Pain scores greater than 5/10 require prompt management (unless this represents a background chronic pain condition where improving the score is unlikely to be achievable).

For patients who may be unable to effectively communicate their pain due to cognitive impairment, the Abbey scale is more useful (also found in the pain assessment tool). However, the Abbey scale does not distinguish between pain and anxiety, so both should be considered.

Pain assessment needs to be performed at least FOUR times daily. Results must be documented, either electronically or using 'comfort round' charts. If 'as required' analgesics are administered, then the patient should be reassessed (and documented) within 1 hour.

Analgesia should be titrated carefully to a point where the pain is acceptable to the patient and facilitates appropriate movement. If this cannot be achieved with usual doses of drugs, advice should be sought from the pain team or anaesthetic teams.

Pain assessment must consider the difference between 'static' pain where the patient is lying still and 'dynamic' pain in which the patient is moving about. Analgesia must enable an appropriate level of movement such as deep breathing, coughing, wriggling in bed and sitting up, as required. When performing painful tasks such as rolling a patient or physiotherapy session, the need for analgesia should be pre-empted and administered sufficiently in advance to allow it time to take effect.

Sedation levels should also be noted, especially when using opioids. Consider the cause of the pain and assess whether a severe worsening of pain also represents worsening of pathology. The presence of nausea and / or vomiting should be recorded.

A respiratory rate of <8 per minute with a deep level of sedation suggests opioid overdose. A slow respiratory rate (8 or above) without over sedation is acceptable - but requires extra vigilance.

5.2. Drug Therapy – General Principals

Consider whether local anaesthesia such as a nerve block can better treat the pain.

Consider whether other methods can be used to ease the pain such as: splinting of fractures, elevation, heat or cold, gentle massage, physiotherapy and focussed breathing for back spasm.

Assess how much distress is contributing to a patient's pain experience and take measures to address this empathically.

If advanced analgesia techniques are being used (e.g. epidural, nerve block or PCA opioids) contact the Acute Pain Management Team or 'on call' anaesthetist if problems arise.

If patient requires analgesics for pain they should be:

- Given by the oral route where possible
- Prescribed on a regular basis for 'baseline' analgesia
- Add 'as required' analgesia to allow for fluctuations in pain intensity
- Titrated until the pain is appropriately managed
- Reducing over time

Before prescribing, check the patient's status with respect to allergy and sensitivities. Use generic drug names only, e.g. 'Oral Morphine Solution' rather than 'Oramorph'.

Avoid the use of combination drugs containing two analgesics, e.g. co-codamol, co-dydramol. Instead prescribe paracetamol and codeine separately.

When prescribing oral solutions, it is important to consider the total volume of a calculated dose and round to the nearest measurable dose in 0.5 ml increments. Example: Ibuprofen = 100 mg / 5 ml, therefore, child of 23 kg = 172.5 mg = 8.6 ml which can be rounded to 8.5 ml (170 mg). With paracetamol the higher concentration (250 mg / 5 ml) which is usually considered for age 6+ may be used in younger children to decrease the volume required and should be considered in the drug calculation.

If the patient is requiring frequent 'as required' opioid analgesia (>3-4 times per day), consider using a low dose regular prescription of a longer acting preparation. Start with codeine or tramadol but sometimes SR morphine or oxycodone is appropriate.

Sustained release opioids require daily review for dose reduction and must be time limited. The decision to start them is best discussed with the pain team and needs to involve a senior doctor to avoid patients inappropriately remaining on long acting opioids.

In acute pain, sustained release morphine and oxycodone should not be used in the manner recommended by the BNF which is for progressive cancer pain. Instead, only ¼ to 1/3 of the previous daily dose should be converted to long acting to allow for fluctuating intensity and reducing need as healing occurs.

Any on-going opioids or newly initiated anti-neuropathic medication MUST be communicated with the patient's GP on discharge. Patients need to be given a plan for dose reduction at discharge and advised that on-going medication (especially opioids) may affect their ability to drive. Further details are found on-line at https://www.gov.uk/drug-driving-law.

Before using NSAIDs consider carefully whether they are appropriate for your patient (paragraph 5.4). Unless contra-indicated they provide important opioid sparing benefit.

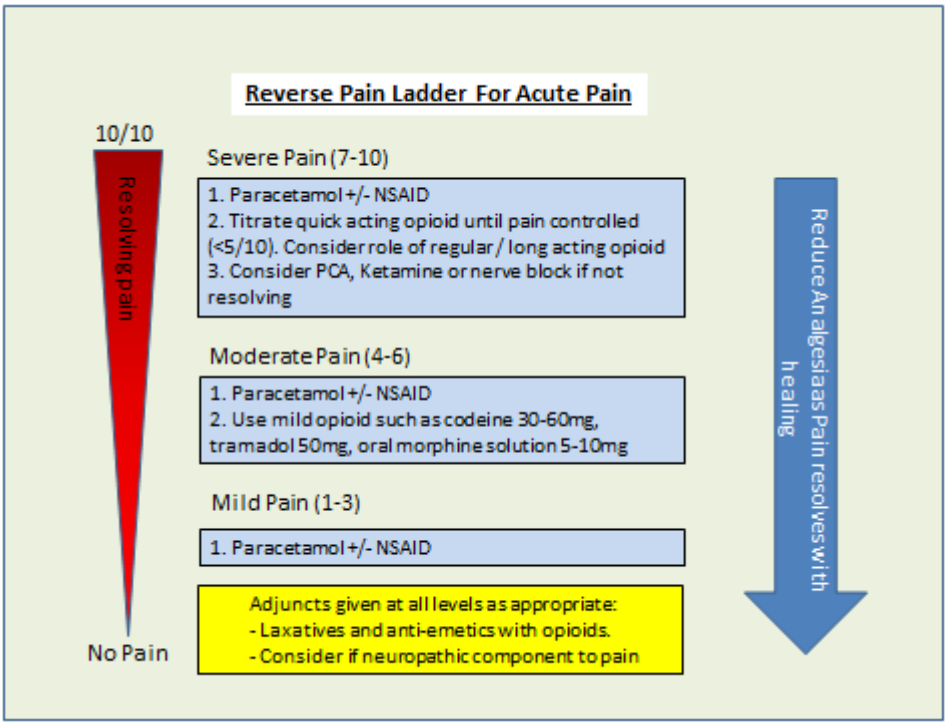
Morphine is the first line choice of opioid. In patients who are allergic to morphine, or who suffer unpleasant side effects such as nausea / vomiting (despite IV anti-emetics), oxycodone or fentanyl (PCA) can be used as an alternative. In patients with significant renal impairment see paragraph 5.7.

All drugs recommended for prescription are done so on the assumption that there are no contra-indications to their use in the individual patient concerned. The BNF should be consulted when initiating any drug which you are not familiar with regarding side effects, indications, doses and contra-indications. If there is any confusion, consult the pain team for clarification and consultant advice if needed.

5.3. Acute Pain – Reverse Analgesic Ladder

The basic principle of using the 'reverse' pain ladder is that pain is generally at its worst immediately following surgery, trauma or acute (painful) illness and improves as tissue damage resolves. However, pain can vary in intensity during that course due to numerous factors such as anxiety, movement, etc. If overall analgesic use is not reducing, a persistent pain state may be developing and the pain team should be informed to assess the patient.

Therefore, analgesic medication must start high (depending on the patient's self-reported severity) and titrated, aiming for a pain score of 4/10 or less. Do not aim to titrate pain scores to 0/10 with opioids as the risk of opioid toxicity will be greatly increased. Regular analgesia must be subsequently offered, with break-through medication 'as required'. As pain resolves, doses can then be reduced and stopped. Adjuncts are given at all levels as appropriate.



5.4. Treatment of Mild Pain: *Paracetamol + NSAIDS*

PARACETAMOL

Paracetamol is classed as a mild analgesic and anti-pyretic due to central prostaglandin inhibition. It can be administered orally, intravenously (IV) or rectally. However, since the introduction of IV preparations, rectal use has decreased markedly in favour of IV.

Adult Dosing (>50 kg):

Route	Dose	Frequency	Maximum Daily Dose
Oral	1g	4-6 Hours	4g
IV	1g	4-6 Hours	4g
Rectal	1g	4-6 Hours	4g

Adults <50 kg or risk of hepatotoxicity:

If patients are at risk of hepatotoxicity due to: weight <50 kg, malnutrition, liver disease or dehydration - reduce the dose. Particular care must be taken with IV

administration to ensure overdose is not inadvertently administered. Either decrease the dose to 750 mg or 500 mg for oral dosing, or prescribe by weight for IV / PR administration:

Route	Dose	Frequency	Maximum Daily Dose
Oral (40-50 kg)	750 mg	4-6 Hours	3g
Oral (<40 kg)	500 mg	4-6 Hours	2g
IV (<50 kg)	15 mg / kg	4-6 Hours	60 mg / kg
Rectal (< 50kg)	40 mg / kg loading 15 mg / kg on going	4-6 Hours	75 mg / kg

Children's dosing:

See Appendix: 'Paracetamol Dosing in Children'.

NSAIDs

Ibuprofen is the preferred choice of NSAID. It is effective with less severe gastric and cardiac side effect profiles than diclofenac. Ibuprofen and naproxen are the NICE recommended NSAIDS. Doses advised by NICE are lower than the BNF maximum dose, but in low risk patients a short course of 600mg, 6-8 hourly may be appropriate:

Route	Dose Ibuprofen	Frequency
Oral (>50 kg)	400 mg	6-8 hourly (Recommend max 1200 mg / day but maximum BNF dose 2400 mg / day)
Oral (< 50 kg)	7.5 (5-10) mg / kg	6-8 hourly (max 30 mg / kg / day)

Children being discharged with ibuprofen should be advised to follow dosing instructions on the bottle.

Naproxen is likely to be the NSAID selected for patients on long term NSAIDs initiated in primary care due to it having the lowest relative risk of cardiovascular events. Dose:

Route	Dose Naproxen	Frequency
Oral (>50 kg)	250-500 mg	1-2 times daily (max 1000 mg per day)

Diclofenac may be used in adults and children when analgesic benefit justifies the increased side effects risk. Suppositories may be used in cases such as after Caesarean Section or children who may struggle to take oral ibuprofen. Dose:

Adults (>50Kg):

Route	Dose Diclofenac	Frequency
Oral	50 mg IR 75 mg SR	8 hourly (max 150mg per day) 12 hourly (max 150mg per day)
Rectal Suppository	100 mg	16 hourly (max 150 mg per day)

Adults (<50Kg) and children (over 6 months / 8 kg):

Route	Dose Diclofenac	Frequency
Oral	1.5 mg / kg	Twice daily
Rectal (8-11 kg)	12.5 mg	2 times daily for maximum 4 days
Rectal (>12 kg)	1 mg / kg	3 times daily (max 150 mg per day) for maximum 4 days

If ketorolac IV (up to 30mg) has been administered during anaesthesia then 6 hours should be left before administering any further NSAIDs.

NSAIDs carry a significant side effect profile and must only be prescribed in appropriate patients. The following list is not exhaustive and care should be taken with all patients that have significant co-morbidities.

Side Effect	Clinical Concern	Recommendation
Gastric Erosion	1. Higher risk if not eating 2. Gastric tubes increase risk further	1. Avoid in patients with NG tube 2. Use with caution if no oral intake 3. Consider using a proton pump inhibitor whilst on NSAID if risk factors
Renal Vasodilatation	1. Can precipitate AKI 2. Hypovolaemia greatly increases risk of AKI	1. Avoid in hypovolaemia 2. Avoid in patients with pre-existing renal impairment 3. Avoid in patients at risk of AKI
Platelet Inhibition	1. Increased bleeding in coagulopathic patients 2. Risk of bleeding in head and spinal injury in particular	1. Avoid in patients with major haemorrhage 2. Avoid in head or spinal injury 3. Avoid in patients with concerns of on-going bleeding
Broncho-constriction	1. Up to 20% of Asthmatics get worsening of symptoms	1. Fine to use if patient knows they are not susceptible 2. Avoid in those known to worsen with NSAIDs 3. Use caution if unknown response and mild asthma. Avoid in brittle asthma.

Do not use in pregnancy.

NSAIDS and **bone healing** has become a controversial issue. The evidence base for this is limited, but there is enough evidence to be concerned about NSAID use in fractures where there is a fear on non-union, especially in smokers. In cases where there may be concern, discussion between the surgeon and doctor prescribing analgesia must ensue as other analgesic techniques can be deployed instead.

5.5. Moderate Pain: *Paracetamol +/- NSAIDS + low dose opioid*

Use regular paracetamol and NSAIDs in the doses described for mild pain.

In addition, a small dose of opioid should be added. Typical for this are the 'weak' opioids codeine or tramadol. Alternatively, a small dose of 'strong' opioid (such as oral morphine solution or oxycodone) may be used.

This should be 'as required' initially. If analgesia is requested frequently, then regular administration should be prescribed with additional opioids 'as required'.

Opioid side effects include nausea, vomiting, constipation, sedation and respiratory depression. Therefore, patients should be prescribed anti-emetics and laxatives as a matter of routine. Naloxone should be used with caution in cases of opioid toxicity as per section 10.

CODEINE

Codeine is not ideal as a drug, but is the most easily administered opioid as it does not have to be treated as a full controlled drug.

It is converted to morphine in the liver with around 10% of the dose of codeine ending up as morphine (i.e. 60mg of codeine is deemed roughly equivalent to 6 mg of oral morphine). However, there is considerable genetic variation affecting this metabolism with 10% of the UK population gaining little benefit. Conversely, some patients are hyper-metabolisers and can develop opioid toxicity from a single dose. Codeine is therefore an unreliable way of giving a small dose of morphine. It also causes constipation.

Codeine is not to be used in children under 12 years of age. It is contra-indicated in children with tonsil induced obstructive sleep apnoea, any respiratory compromising condition or known CYP2D6 hyper-metabolisers. Breast feeding mothers should also avoid codeine as it goes into the milk.

Dose:

Route	Dose	Frequency
Oral (>12 years)	30-60 mg	4 times daily (max 240 mg per day)
Oral (Elderly / frail)	15-30 mg	4 times daily

Tramadol

Tramadol has noradrenaline and serotonin effects as well as being an opioid analgesic. This gives it extra analgesic mechanisms compared to codeine and whilst also subject to genetically variable liver metabolism, it gives a more predictable analgesic effect. However, these extra effects are associated with significant side effects, making many patients feel 'drunk'.

This is particularly apparent in the elderly, in whom tramadol should be used with caution, so it is generally reserved for those who are known to tolerate it well or are already using it for chronic pain conditions. Tramadol should also be avoided in patients with epilepsy and caution used with patients concurrently taking mono-amine

oxidase inhibitors (MAOI), serotonin specific re-uptake inhibitors (SSRI) or Tricyclic anti-depressants.

Dose (IR):

Route	Dose	*Frequency*
Oral (>12 years)	50-100 mg	*4 times daily (max 400 mg per day)*

Unless already using in the community, do not prescribe Tramadol SR without discussion With the Pain Team.

Low Dose 'Strong' Opioid

Some patients may find that it is best to bypass codeine / tramadol and use a small dose of morphine or oxycodone directly for moderate pain. As always, this should be titrated to effect.

For example, if a patient is known to not respond well to codeine or tramadol, then oral morphine solution 5 mg or oxycodone IR 2.5 mg (allowing for age, renal function, etc) may be appropriate as an alternative. This should be prescribed 'as required' with 2-4 hours between doses but considered regularly on each drug round.

If this is inadequate then carefully increase the 'as required' dose, eg oral morphine solution 5-10 mg or oxycodone IR 2.5-5 mg with 2-4 hours between doses. Again, adjust dose for age, renal function, etc. Typical doses:

Drug	Route	Dose	Frequency
Oral Morphine Solution	Oral	5-10 mg	2-4 Hours 'As required'
Oxycodone	Oral	2.5-5 mg	2-4 Hours 'As required'

5.6. Severe Pain: *Paracetamol +/- NSAID + titrated dose 'strong' opioid*

Use regular paracetamol and NSAIDs (where appropriate) in the doses described for mild pain.

STRONG OPIOIDS

In addition, 'strong' opioids should be added, titrated until the pain is at an acceptable level with pain scores of 4 or lower. Do not focus on trying to titrate to a pain score of 0/10 as this will induce an unnecessarily high risk of opioid toxicity.

Morphine is the first line opioid in most cases. For patients who are allergic or intolerant of morphine, or have significant renal impairment, alternative opioids such as oxycodone or fentanyl (PCA only) should be considered. Reduce dose of morphine if eGFR <50 ml / min and avoid if eGFR <30 ml / min (See paragraph 5.7).

Fentanyl lozenges are not to be used for acute pain. Fentanyl patches in the acute setting are not to be used unless prescribed by a Consultant Anaesthetist.

'Strong Opioid' doses depend on age, renal function, previous opioid exposure and sensitivity; as well as degree of pain experienced. Therefore, these doses are a guide only and must prescribing must take into account individual factors. In the elderly, lower doses are generally required. In renal impairment adjust doses as per paragraph 5.7. Opioid tolerant patients may need up to double these doses on advice of the Acute Pain Team.

Adult typical dose ranges for ward use:

Morphine:

Patient Factors	Route	Dose	Frequency
Frail / Elderly / Opioid sensitive	Oral	5-10 mg	2-4 Hours 'As required'
Frail / Elderly / Opioid sensitive	IV / IM / SC	2.5-5 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	Oral	10-20 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	IV / IM / SC	5-10 mg	2-4 Hours 'As required'

Oxycodone:

Patient Factors	Route	Dose	Frequency
Frail / Elderly / Opioid sensitive	Oral	2.5-5 mg	2-4 Hours 'As required'
Frail / Elderly / Opioid sensitive	IV / IM / SC	2.5 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	Oral	5-10 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	IV / IM / SC	2.5-5 mg	2-4 Hours 'As required'

Morphine Doses in Children:

Route	Age	Dose	Frequency
Oral	<6 Months	50-100 mcg / kg	4 Hourly (Adjusted to response)
Oral	6-12 Months	200 mcg / kg	4 Hourly (Adjusted to response)
Oral	1-12 Years	100-300 mcg / kg	4 Hourly (Adjusted to response)
Oral	12-17 Years	5-10 mg	4 Hourly (Adjusted to response)

Route	Age	Dose	Frequency
IV	1-5 Months	100 mcg / kg	6 Hourly (Adjusted to response)
IV	6 Months – 11 Years	100 mcg / kg	4 Hourly (Adjusted to response)
IV	12 -17 Years	5 mg	4 Hourly (Adjusted to response)

These doses are typical ranges and both higher and lower doses may be used by experienced doctors. Other regimens such as Intranasal diamorphine in the Emergency Department outside of this guideline and should be used by local protocol.

Sustained Release Opioids (Morphine SR and Oxycodone SR)

If an adult patient is requesting frequent opioid administration, (or can be anticipated to do so) a regular form of opioid should be considered. This may be in the form of regular codeine or tramadol, but SR morphine or SR oxycodone may be used. Sustained release opioids in children are

When using sustained release morphine or oxycodone, there is no benefit conferred by using codeine in addition, but tramadol may be still of benefit due to its extra analgesic effects.

If patients are requiring regular oral opioids for an acute condition, then long acting preparations can be used. This is not done in the manner recommended by the BNF which applies to the standard, progressive WHO pain ladder for cancer pain.

For acute pain, using the reverse ladder, no more than 1/3 of the previous day's total opioid dose should be converted to long acting. Example:

Previous day 6 x 20mg oral morphine solution = 120 mg
Convert 1/3 to long acting morphine = 20 mg BD
Continue PRN oral morphine solution at 10-20 mg, max 2 hourly
Expect patient to then request up to 3-4 doses of oral morphine solution per day

On occasion, an anaesthetist may prescribe SR morphine or oxycodone. This is sometimes done to smooth out the delivery of opioid in the first 24-48 hours following moderate or major surgery. In such cases this needs to be explained to the patient and the number of doses explicitly detailed in the prescription.

It is MANDATORY that any SR morphine / oxycodone initiated for *acute* pain is time limited (eg 3 days only) and reviewed daily to reduce and stop it as the need decreases. Patients being discharged with on-going opioids need review by the pain team and their GP notified explicitly on the discharge summary.

Patients must also be informed that these drugs may impair their ability to drive. Patients can be directed to the DVLA website (https://www.gov.uk/drug-driving-law) for further guidance on the law regarding this.

All opioids should be prescribed with caution and at reduced doses in patients with histories of:

- Renal or hepatic dysfunction
- Respiratory dysfunction
- Obesity / Sleep apnoea syndromes
- Elderly

In patients with a reduced level of consciousness, opioids should not be used at all except by a specialist, e.g. Pain Management Specialist, Palliative Care Specialist, Oncologist or Anaesthetist.

5.7. Opioids in Renal Impairment

Morphine

Avoid sustained release (SR) preparations in renal impairment.

eGFR (ml / min)	Dose Change	Time Interval
50-90	No change	2 Hours
30-50	Max 50-75% usual	4 Hours
<30	Do not use	Do not use

Codeine / Dihydrocodeine

eGFR (ml / min)	Dose Change	Time Interval
50-90	No change	4-6 Hours
30-50	Max 30mg initially Can increase if tolerated	4-6 Hours
<30	Do not use	Do not use

Tramadol

eGFR	Dose Change	Time Interval
50-90	No change	6 Hours
20-50	No change	8 Hours
10-20	50 mg	8 Hours
<10	Do not use	Do not use

Oxycodone

Oxycodone is safer than morphine and should be used preferentially over morphine in patients with an eGFR <30 ml / min. It is still affected by renal impairment, therefore use only IR preparations and adjust accordingly:

eGFR	Dose Change	Time Interval
60-90	No change	2 Hours
30-50	Max 50% usual initially Increase if tolerated	4 Hours
10-30	Max 50% usual initially Increase if tolerated	6 Hours
<10	Seek advice	Seek advice

Fentanyl

Fentanyl is the only ‘safe’ opioid in renal impairment, but can only be used in IV form for acute pain – usually PCA. However, consider reducing the dose to 50-75% usual initially.

Fentanyl lozenges are not to be used without pain team approval, unless the patient already uses them in the community or is suffering with break-through cancer pain (contact palliative care team).

Pethidine

Pethidine's use is not encouraged by this guideline for general use and it has a limited role outside of obstetrics. Avoid using pethidine in any renal impairment.

5.8. Opioid Dose Equivalents

Morphine has variable potency when administered via different routes, due to altered rates of absorption into the blood and hepatic metabolism when delivered via the gut.

It is not acceptable to prescribe opioids as PO/IV/IM in a single prescription as these different routes have different potencies and time of action (see Morphine potency table below). Instead, different routes of administration should be prescribed separately and annotated to ensure that both routes are not used simultaneously.

Morphine Potency Table			
Route	**Potency**	**Time of action**	**Typical Doses**
PO	**X1**	**30-45mins**	**20mg 2 Hourly PRN**
SC	**X2**	**10-20mins**	**10mg 4 Hourly PRN**
IM	**X2**	**10-15mins**	**10mg 4 Hourly PRN**
IV	**X4**	**3-5mins**	**5mg 2-4 Hourly PRN**

5.9. Drugs for Neuropathic Pain

Patients with pain not responding to standard analgesia may have neuropathic pain. This is pain arising from within the nervous system itself. Examples are pain following peripheral nerve damage, phantom limb pain, post stroke pain or sciatica. It is notoriously difficult to treat and generally poorly responsive to opioids.

Consider neuropathic pain when pain exceeds expectation given the amount of tissue damage, the surgery or trauma may include nerve trauma (eg limb amputation) and the patient describes symptoms such as burning, stabbing pain or paraesthesia / electrical sensations.

If neuropathic pain is suspected, merely increasing opioids is unlikely to be helpful. The pain team should be informed and can advise of treatment options. In the immediate post-operative phase, regional nerve blockade or a ketamine infusion may be appropriate following assessment by an anaesthetist.

Starting anti-neuropathic medication such as pregabalin and gabapentin have long term implications and must be communicated with the patient's GP for review as the dose may need increasing or the drug ceased due to side effects, symptom

improvement or lack of benefit. It must be noted that probably only around 1 in 3 persons initiated on pregabalin and gabapentin will benefit enough (with tolerable side effects) to justify continuing in the long term.

Patients should have an informed discussion about side effects such as dizziness, blurred vision, drowsiness and weight gain to consent to initiating therapy. Therefore, the pain team should be involved in decisions to do so. Patient information leaflets are advised to be used:

https://www.britishpainsociety.org/static/uploads/resources/files/FPM-Pregablin_2.pdf
https://www.britishpainsociety.org/static/uploads/resources/files/FPM-Gabapentin_0.pdf
https://www.britishpainsociety.org/static/uploads/resources/files/FPM_Amitriptyline.pdf

Patients must also be informed that these drugs may impair their ability to drive and direct patients to the DVLA website (https://www.gov.uk/drug-driving-law) for further guidance..

The patient's GP must be informed and a plan for continuation and review must be in place for discharge.

PREGABALIN

This is an anti-convulsant which blocks neuronal calcium channels to dampen down neuronal activity. Around one person in every 3 will gain enough reduction in symptoms and tolerable side effects to justify remaining on pregabalin long term.

If side effects outweigh benefit then a dose reduction must be considered or stopped (if of no benefit). Doses should be reduced gradually to avoid precipitous withdrawal. It is also expensive and should generally be used as a second line drug to its predecessor, gabapentin.

However, due to having a better side effects profile then gabapentin, it can be titrated much more quickly making it useful in the acute setting. It should not be initiated on patients without discussion with a pain consultant first.

Side effects are generally of drowsiness, dizziness, blurred vision and long term can cause significant weight gain (for full list and contra-indications see BNF). Patients need to have a discussion of the risks and benefits for informed consent prior to initiating therapy. The patient's GP must be informed and a plan for continuation and review must be in place for discharge.

Contra-indications and cautions include conditions precipitating encephalopathy and severe congestive heart failure.

Dose (according to BNF) adults only, use in children is not covered by this guideline:

Route	Dose	Frequency
Oral (Week 1)	75 mg	BD (roughly 12 hourly)
Oral (Week 2)	150 mg (working dose)	BD (roughly 12 hourly)
Oral (Week 3)	300 mg (high dose)	BD (roughly 12 hourly)

Do not use Pregabalin in pregnancy unless authorised by a consultant and risks agreed by the patient.

Many patients may find side effects at the above doses prohibitive and a lower dose with slower increases is often necessary, eg starting with 50 mg BD and increasing to 100 mg BD after a week.

Conversely, some patients can tolerate a more rapid build-up of dose when in-patient and suffering acute neuropathic pain. Rapid dose increases are to be performed only on the advice of a pain consultant.

Pregabalin undergoes renal excretion and the doses must be reduced in renal impairment:

eGFR	Initial (oral) Dose	Frequency
>60	75 mg	BD (max 600 mg per day)
30-60	75 mg	OD (max 300 mg per day)
15-30	25 mg	OD (max 150 mg per day)
<15	25 mg	OD (max 75 mg per day)

GABAPENTIN

This works in a similar manner to pregabalin but is much cheaper and of comparable efficacy. It is also reasonably tolerated in many patients. Therefore, it should be considered as first line of the two, when circumstances permit.

The main advantages of pregabalin over gabapentin are that pregabalin is administered twice daily (compared to three times daily with gabapentin) and that it has a better side effect profile. Pregabalin can also be titrated more rapidly.

In select in-patients gabapentin can be titrated in an accelerated manner, but only under the auspices of a pain consultant.

Dose (adult only, use in children is not covered by this guideline):

Route	Dose	Frequency
Oral (day 1)	100 mg	Three times daily
Oral (day 2)	200 mg	Three times daily
Oral (day 3)	300 mg	Three times daily
Oral (day 4)	400 mg	Three times daily
Oral (day 5)	500 mg	Three times daily
Oral (day 6)	600 mg	Three times daily

This varies slightly from BNF doses, but uses the same daily dose at each point and is commonly accepted as a smoother way of titrating doses. For titration in the out-patient setting, the dose interval is usually changes to weekly, rather than daily.

Side effects are generally of drowsiness, dizziness, blurred vision and long term can cause significant weight gain (for full list and contra-indications see BNF). Patients

need to have a discussion of the risks and benefits for informed consent prior to initiating therapy. The patient's GP must be informed and a plan for continuation and review must be in place for discharge. Do not use in pregnancy unless authorised by a consultant and risks agreed by the patient.

Apply caution in diabetes mellitus; elderly (in adults); high doses of oral solution in adolescents and adults with low body-weight; history of psychotic illness; mixed seizures (including absences).

Renal impairment can greatly increase the effects of gabapentin, making dose reduction necessary (BNF doses):

eGFR	Maximum Dose	Frequency
50-80 ml / min	200-600 mg	Three times daily
30-50 ml / min	100-300 mg	Three times daily
15-30 ml / min	100-200 mg	Three times daily
<15 ml / min	100 mg	Three times daily (alternate days)

AMITRIPTYLINE

This is a tri-cyclic antidepressant which can be of benefit in neuropathic pain. Although it is not licenced for neuropathic pain, NICE guidelines support its use and many formularies advocate it as first line, usually due to its very low cost.

The dose is 10 mg around an hour before bed. This can be increased to 20 mg after a few days if well tolerated. Side effects include dry mouth, drowsiness and difficulty with micturition, especially in men with enlarged prostates (for full list and contra-indications see BNF).

Contra-indications: Acute porphyrias; arrhythmias; during manic phase of bipolar disorder; heart block; immediate recovery period after myocardial infarction. Do not use in pregnancy unless authorised by a consultant and risks agreed by the patient.

Apply caution in cardiovascular disease; chronic constipation; diabetes; epilepsy; history of bipolar disorder; history of psychosis; hyperthyroidism (risk of arrhythmias); increased intra-ocular pressure; patients with a significant risk of suicide; phaeochromocytoma (risk of arrhythmias); prostatic hypertrophy; susceptibility to angle-closure glaucoma; urinary retention. However, at the low doses used for neuropathic pain its use may still be appropriate.

5.10. Adjuncts

Adjuncts are all the things that make analgesia work better. As all opioids cause nausea and constipation, anti-emetics and regular laxatives must be prescribed. Ideally, laxatives should have softening and motility functions. Ondansteron, cyclizine and prochlorperazine can be useful anti-emetics and should be used in accordance with the BNF.

6. Acute Pain Relief in Opioid Dependent Patients

6.1. Introduction

These patients may be using opioids due to drug dependency or prescribed due to persistent pain. They generally fall into one of four groups:

- Unknown quantities of street opioids
- Regular oral methadone (substitute for street drugs or for persistent pain)
- Regular oral opioids for persistent pain
- Regular opiate patches for persistent pain

Pain in these patients can be difficult to manage, therefore contact the Acute Pain Management Team early.

However, ward teams need to confirm the doses of any patient on high dose opioids with the patient's GP, community pharmacy or local drug addiction service, as appropriate. This 'background' dose should be continued and extra analgesia prescribed in addition.

6.2. Essential principles of practice

Medical and surgical teams must prescribe a background opioid prescription that replicates / replaces the opioids taken by the patient on a regular basis. This is given the term 'basal opioid'. If a patient is unable to take this orally, then a parenteral substitute must be provided. This is likely to be IV and may well be in the form of a PCA system.

Add a regular component for the acute pain – 'regular acute pain dose'. Use non-opioid techniques for analgesia where ever possible. Paracetamol IV is a useful starting point and consideration to using regional anaesthesia techniques should be given.

Increasing the regular dose of long acting opioid may be appropriate, but must be monitored and returned to base line by discharge. Any on-going increases in opioids at discharge must be communicated with the patient's GP with a plan for further reductions. There is no benefit in adding codeine regularly in these patients, but tramadol may be of value due to its non-opioid effects. Do not increase methadone doses.

'As required' opioid doses may need to be significantly higher than normal. For example, the dose of oral morphine solution could be as high as 20-40mg, 2 hourly PRN. A low threshold for utilising PCA opioid may enable safer titration of doses. Only use morphine, oxycodone or fentanyl, unless advised otherwise by a Consultant Anaesthetist.

6.3. General Principles of Practice in Drug Dependence

Each case needs to be planned individually.

Confirm opioid doses with an independent source e.g. GP, local pharmacist, drug service.

Morphine, oxycodone and fentanyl are good opioids to use. Pethidine is less versatile and in the large doses often required in these patients can cause fits (neurotoxic metabolite accumulation).

People abusing drugs other than opioids, e.g. amphetamines, cocaine - treat as non-dependent. Cross sensitivity is unusual although sedative effects can be additive.

Monitor the patient carefully with hourly observations for the first 24 hours and beyond if necessary.

The prescription often needs to be fine-tuned and large doses are often needed. Usually complications are rare but unreliable histories, illicit use and variable tolerance to the side effects are possible and monitoring is mandatory.

If patients are presenting for minor/day case surgery then it is acceptable to leave them on their basal opioid which they should take as normal and add in oral analgesia as appropriate.

6.4. Special Considerations

Persistent pain patients

These patients can present taking a variety of opioids, which should not be stopped without consultation. Some patients with persistent pain may have opioid dependant types of behaviour. Whilst this may represent an unhelpful manner of using opioids, it may be due to the nature of their pain and should not incur prejudicial withholding of analgesia.

High doses of opioid use (>120mg oral morphine equivalent per day) are now deemed to cause more harm than benefit in the chronic setting and can, in fact, sensitise patients to pain rather than relieve it.

Therefore, individual and non-judgemental assessment must be made as to the most appropriate way of treating their pain. Look for a pain plan in the notes if there have been multiple admissions due to pain. Opioids may be appropriate but also can exacerbate long standing problems e.g. by making patients with chronic abdominal pain constipated. Early referral to the pain team and caution with opioids is required. Non-opioid ways of managing pain must be optimised. The acute setting is not the time to withdraw opioids, unless under specialist guidance.

Patients with Previous History of Drug Misuse

If these patients are not taking methadone then they should be treated as other patients with no alterations in analgesic doses. There is no evidence that using opioids for pain relief precipitates a relapse into drug abuse. However, use a local anaesthetic / regional technique or the oral route whenever possible. When IV opioids are necessary, using a PCA can give a smoother profile of administration than intermittent IV boluses.

6.5. Acute pain relief for patients with advanced disease who are already taking opioids - general principles

Refer to: Symptom management in the Palliative Care.

7. IV or SC Morphine / Oxycodone Administration on the Ward

7.1. Intravenous Guidelines

Intravenous morphine or oxycodone should be used for the initial control of acute severe pain only. Nurses giving IV opioids must have completed the Trust's IV drug training and assessment package. IV doses of morphine and oxycodone are identical, subject to renal impairment alterations as described in paragraph 5.7.

- Ensure IV morphine (or oxycodone) has been correctly prescribed.
- Make up a syringe containing 10 mg of morphine **or** oxycodone made up to 10mls with 0.9% sodium chloride, following Trust policy.
- If the patient has not had any previous opioid, give 2.5 – 5mg (of either medicine) IV over 1 minute and repeat if still in pain after 5 minutes. If the patient requires larger doses, consider PCA and Consult on-call anaesthetist or acute pain team.
- During injection observe the cannula site for extravasation or local sensitivity reactions. Mild reactions can occur after administration of IV morphine due to localised histamine release. Significant redness or swelling should be reviewed by a doctor and may represent an allergic response.
- Monitor the patient's conscious level, blood pressure, pulse, respiratory rate and oxygen saturation every 5 minutes AND for 20 minutes after the last dose of morphine / oxycodone has been injected.

Note: Patients with cancer may already be on large doses of opioids. Please contact Palliative Care, Oncology or Pain Team for advice as the intravenous doses may need to be larger. You can contact the Palliative Care team on bleep 401.

7.2. Subcutaneous Morphine Injection on the Ward

Guidelines:

See Symptom management in the Palliative Care.

8. Patient Controlled Analgesia (PCA)

8.1. Guidelines

The system consists of a syringe pump, filled with opioid, connected to an IV cannula via a narrow bore giving set. The pump is operated by a button, which the patient presses, to administer a small dose of drug, the bolus dose. After the bolus dose has been given (usually over 1 minute), the system will not respond for a pre-set time, the 'lock-out' time. For PCA this is usually 4 minutes.

This allows the patient to feel the effect of the drug before receiving a further dose, thus preventing overdose but permitting a bolus every 5 minutes. The dose and the lockout period are pre-set according to the protocols but can be varied within the pre-set range.

The nurse caring for the patient must be a registered nurse who has completed a training package, or be supervised by a registered nurse who is competent in the care of patients receiving PCA.

Patient selection is normally by the anaesthetist, who will provide the prescription. Suitable patients will have both the physical and cognitive capacity to both understand how to use the system and be physically capable of pressing the button.

Only the patient may press the button. Nurses and relatives and doctors are not covered by the prescription to administer boluses via the button and constitute a non-prescribed administration of opioid in doing so. Clinician administered boluses can be delivered as a separate function of the pump by pain team and anaesthetic staff covered by an additional prescription.

Pre-operative education is desirable and patients should be given the Patient Information leaflet on PCA. Nursing staff on the ward and in Recovery will reinforce this education.

At the end of surgery, a pre-filled syringe will be loaded into the system, with the pump programmed and locked. If the PCA infusion is connected to an existing infusion system, a one-way valve must be used in the fluid line (not in the morphine line). PCA must be used with a specific designated line with an anti-syphon and anti-reflux valve.

Naloxone and anti-emetic medication should also be prescribed.

Other opioids must not be given whilst the PCA is in use, unless advised by an anaesthetist; this includes codeine, tramadol & Dihydrocodeine. Prescribed night sedation and non-opioid analgesics can be given if required.

Keys from the machine will be kept on the clinical areas controlled drug keys. Recovery unit will keep 4 sets of machine keys. Ward nurses are NOT allowed to carry their own set of PCA keys. Syringes and lines are to be changed in accordance with the Trust guidelines.

Specific nursing observations are essential - refer to PCA prescription charts (see Appendix C, Appendix D). If the respiratory rate falls below 8 breaths per minute or the patient is difficult to rouse, remove the control button from the patient and give oxygen and naloxone as needed.

If analgesia is inadequate, check the cannula site to ensure that it is patent and that the drug is being delivered. If there is no mechanical cause, give other non-opioid prescribed analgesics. If pain continues to be moderate or severe on movement, call the Pain Team or on-call anaesthetist. It may be necessary to adjust the pump programme.

Prior to stopping the PCA, ensure that the patient has been prescribed suitable alternative analgesics and that these are sufficient for the current level of pain.
.
Note: The pump programme can be set or altered by the anaesthetist, the Pain Team and nurses who have had additional PCA training in pump programming.

8.2. Prescription Options for Patient Controlled Analgesia (PCA) for Adults and Children >50kg

- **Morphine:** 50mg made to 50ml with normal saline (1 mg / ml).
  - Bolus 0.5 mg (0.5 ml over 1 minute), Lock out 4 minutes.
  - Bolus 1 mg (1 ml over 1 minute), Lock out 4 minutes.
  - Bolus 2 mg (2 ml over 2 minutes), Lock out 3 minutes.
  - Background infusions may only be added for use in Critical Care (1-2 ml / hour) or Caroline Thorpe ward (0.2 ml / hour).

- **Oxycodone:** 50mg made to 50ml with normal saline (1 mg / ml).
  - Bolus 0.5 mg (0.5 ml over 1 minute), Lock out 4 minutes.
  - Bolus 1 mg (1 ml over 1 minute), Lock out 4 minutes.
  - Bolus 2 mg (2 ml over 2 minutes), Lock out 3 minutes.
  - Background infusions may only be added for use in Critical Care (1-2 ml / hour) or Caroline Thorpe ward (0.2 ml / hour).

- **Fentanyl:** 500mcg made to 50ml with normal saline (10 mcg / ml).
  - Bolus 5 mcg (0.5 ml over 1 minute), Lock out 4 minutes.
  - Bolus 10 mcg (1ml over 1 minute), Lock out 4 minutes.
  - Bolus 20 mcg (2ml over 2 minutes), Lock out 3 minutes.
  - Background infusions may only be added for use in Critical Care (1-2 ml / hour) or Caroline Thorpe ward (0.2 ml / hour).

8.3. Prescription Options for Children Weighing <50 Kg

Children weighing less than 50 Kg require a separate prescription using either morphine or oxycodone:

- **Morphine:** 1 mg / Kg made to 50ml with normal saline (20 mcg / Kg / ml).
  - Bolus 0.5 mg (0.5 ml over 1 minute), Lock out 4 minutes.
  - Bolus 1 mg (1 ml over 1 minute), Lock out 4 minutes.
  - Background infusions may only be added for use in Critical Care (1 ml / hour) or Caroline Thorpe ward (0.2 ml / hour).

- **Oxycodone:** 1 mg / Kg made to 50ml with normal saline (20 mcg / Kg / ml).
  - Bolus 0.5 mg (0.5 ml over 1 minute), Lock out 4 minutes.
  - Bolus 1 mg (1 ml over 1 minute), Lock out 4 minutes.
  - Background infusions may only be added for use in Critical Care (1 ml / hour) or Caroline Thorpe ward (0.2 ml / hour).

9. Nurse Controlled Analgesia (NCA)

9.1. Guidelines

Nurse controlled analgesia (NCA) is an option for children who are unable to manage a PCA themselves. Its use is therefore confined to Critical Care and Caroline Thorpe ward.

Adult patients who are deemed unable to manage a PCA should be managed with intermittent boluses of morphine or oxycodone, prescribed 'as required'. This should be administered following an assessment of the patient's pain and appropriate observations performed after 15-20 minutes. Such patients can be managed on routine wards and do not specifically require Critical Care, however, extra nursing staff may be required if frequent boluses are needed.

The system consists of a syringe pump, filled with opioid, connected to an IV cannula via a narrow bore giving set. The pump is operated by a button, which the NURSE presses, to administer a small dose of drug, the bolus dose. After the bolus dose has been given (usually over 1 minute), the system will not respond for a pre-set time, the 'lock-out' time.

This is the same as for PCA, but loses the inherent safety of relying on the patient being sufficiently awake to press the button themselves. Therefore a longer 'lock-out' period of 20 minutes is used, but with a continuous low dose infusion at 1 ml / hour.

The nurse caring for the patient must be a registered nurse who has completed a training package, or be supervised by a registered nurse who is competent in the care of patients receiving NCA.

Only the nurse responsible for the patient may press the button. Relatives and other staff are not covered by the prescription to administer boluses via the button and constitute a non-prescribed administration of opioid in doing so. Clinician administered boluses can be delivered as a separate function of the pump by pain team and medical staff, covered by an additional prescription.

At the end of surgery, a pre-filled syringe will be loaded into the system, with the pump programmed and locked. If the NCA infusion is connected to an existing infusion system, a one-way valve must be used in the fluid line (not in the morphine line). NCA must be used with a specific designated line with an anti-syphon and anti-reflux valve.

Naloxone and anti-emetic medication must also be prescribed. Other opioids must not be given whilst the NCA is in use, unless advised by an anaesthetist. Prescribed night sedation and non-opioid analgesics can be given if required.

Keys from the machine will be kept on the clinical areas controlled drug keys. Recovery unit will keep 4 sets of machine keys. Ward nurses are **NOT** allowed to carry their own set of NCA keys. Syringes and lines are to be changed in accordance with the Trust guidelines.

Specific nursing observations are essential - refer to NCA prescription charts (see Appendix E). If the respiratory rate falls below 8 breaths per minute or the patient is difficult to rouse, remove the control button from the patient and give oxygen and naloxone as needed.

If analgesia is inadequate, check the cannula site to ensure that it is patent and that the drug is being delivered. If there is no mechanical cause, give other non-opioid prescribed analgesics. If pain continues to be moderate or severe on movement, call the Pain Team or on-call anaesthetist. It may be necessary to adjust the pump programme.

Prior to stopping the PCA, ensure that the patient has been prescribed suitable alternative analgesics and that these are sufficient for the current level of pain. Note: The pump programme can be set or altered by the anaesthetist, the Pain Team and nurses who have had additional PCA training in pump programming.

9.2. Prescription Options for Nurse Controlled Analgesia (NCA)

- **Morphine:** 1 mg / Kg made to 50ml with normal saline (20 mcg / Kg / ml).
  - Bolus 1 ml over 1 minute, Lock out 20 minutes.
  - Background infusion 1 ml / hour.
  - Children over 50 Kg in weight use 50 mg in 50 ml.

- **Oxycodone:** 1 mg / Kg made to 50ml with normal saline (20 mcg / Kg / ml).
  - Bolus 1 ml over 1 minute, Lock out 20 minutes.
  - Background infusion 1 ml / hour.
  - Children over 50 Kg in weight use 50 mg in 50 ml.

10. Management of Opioid Side Effects

10.1. Excessive sedation / Respiratory Rate 5-8 per Minute with Adequate SpO2

A patient sedated by opioids to a level where they only respond to vocal commands constitutes excessive sedation. Give oxygen via a face mask. Call doctor. Ensure airway is patent. Try and rouse the patient. If unsuccessful use naloxone:

Adults + Children >12 Years:
Naloxone 100 mcg IV, repeated every 1-2 minutes until patient is responsive.

Children <12 Years:
Naloxone 1 mcg / kg, repeated every 2 minutes until patient is responsive.
(See Naloxone SOP)

Check sedation score every 15 minutes for one hour following the last dose of naloxone. Further doses may be required as naloxone is shorter lasting than morphine.

10.2. Severe Respiratory Depression: <5 Breaths per Minute +/- Low SpO_2

Adults +Children >12 Years:
Naloxone 400 mcg IV, repeated every 1-2 minutes until patient is responsive.

Children <12 Years:

Naloxone 1 mcg / kg, repeated every 2 minutes until patient is responsive. (See Naloxone SOP)

Check respiratory rate every 15 minutes for one hour following the last dose of naloxone. Further doses may be required as naloxone is shorted lasting than morphine.

10.3. Treatment of Nausea and Vomiting

Please refer to the Post Operative Nausea and Vomiting (PONV) Guidelines

Contact Acute Pain Team if problem persists. Generally consider: Hypoxia, hypovolaemia, hypotension, full stomach. If nausea and vomiting persists, a successful drug combination may need to be prescribed on a regular basis. All anti-emetic prescriptions should be reviewed after 48 hrs and the need for continuing prescription assessed.

10.4. Constipation

This is a side effect for most patients taking opioids. Bowel function should be checked and noted at least once daily. Prescribe (adults):

- Docusate sodium 200mg Oral, twice daily.
- Add as necessary, according to local formulary

10.5. Urinary Retention

Consider indwelling urinary catheter.

10.6. Itching

This may occur at the site of injection which needs no treatment.

Generalised itching that is troublesome may be treated with chlorphenamine 4mg PO 8 hourly. If this is not effective try 40-80 micrograms of naloxone given slowly IV. Beware of acutely reversing the analgesia.

Contact the Acute Pain Team if the problem persists.

11. Epidural Analgesia

11.1. Introduction

Epidural injection or infusion, at the lumbar and lower thoracic level, is suitable for most abdominal and lower limb surgery. An epidural catheter is inserted and local anaesthetic, with or without the addition of opioids, is infused through this catheter. The system consists of a narrow-bore catheter with antibacterial filter and a premixed bag containing the prescribed drug(s). The extension line between the syringe and

antibacterial filter is coloured yellow. These extensions should only be used for infusions of local anaesthetic +/- opioids.

Epidurals, paravertebral catheters and nerve catheter infusions must NEVER be connected to an intravenous system.

The Trust has standardised on the following ready to administer infusions for general use:

- Bupivacaine 0.167% in Sodium Chloride 0.9% (500mls)
- Bupivacaine 0.1% with Fentanyl 2microg/ml (500mls)

Obstetrics have a standardised mix of Bupivicaine 0.1% with fentanyl 2microg/ml (100ml bags). Obstetric analgesia protocols may differ slightly from those covered by this guideline.

If the bupivacaine 0.167% solution is used (without epidural opioids) then PCA opioids maybe added in tandem. If the fentanyl containing solution is used NO further opioids are to be administered except on the authority of an experienced anaesthetist.

Any other analgesia for epidural infusions required to treat specific patient (diamorphine, Fentanyl, levobupivacaine 0.25%) should only be prescribed and administered by an anaesthetist. The pain team may also administer a bolus on the (documented) recommendation of an anaesthetist.

The administration of medicines by the epidural route is potentially hazardous and it is important that all those who prescribe, administer and monitor the administration of medicines by the epidural route should be trained and competent.

Due to fatal cases of epidural / intrathecal drugs being administered via the wrong route, the National Patient Safety Agency (NPSA) a number of years ago issued a patient safety alert practice with epidural injections and infusions to use non-compatible connectors:
http://www.nrls.npsa.nhs.uk/alerts/?entryid45=94529.

However, nationally this has yet to be implemented. Until such time, extra care must be taken when administering solutions via epidural / intra-thecal routes to ensure that they are not erroneously given intravenously. A ‘stop before you inject’ moment to pause and confirm the route of administration should be performed. The syringe must also be labelled with a **yellow** label.

11.2. Equipment for local anaesthetic infusions

Equipment for local anaesthetic solutions should never be attached to an intravenous cannula or line. Likewise, solutions for intravenous administration should not be connected to local anaesthetic catheters. To help distinguish between the two systems, equipment used for local anaesthetic solutions is coded **yellow**.

The Body Guard infusion pump and designated giving set are only to be used for epidural infusions or nerve catheter infusions of local anaesthetic.

11.3. Contraindications For Epidural Insertion

Absolute

Unstable Hypovolaemia

Coagulopathies

Local sepsis

Allergy to drugs used

Raised intracranial pressure

Unstable spinal fractures

Patient does not give consent

Relative

Pyrexia or septicaemia

Cardiovascular disorders such as aortic or mitral stenosis

Neurological disorders or Spinal deformity

Hypertension

11.4. Epidural Guidelines

Responsibility

Patient selection is determined by the anaesthetist. The anaesthetist is responsible for obtaining patient consent for the procedure.

Although on-going patient care will be delegated to nursing staff, on-call anaesthetic teams and the Acute Pain Team, legal responsibility for an epidural analgesic system lies with the consultant anaesthetist responsible for its insertion and extends for its entire duration. He / she should be informed of significant complications at the earliest appropriate moment.

The anaesthetist siting the epidural MUST liaise with the receiving ward to ensure that appropriately competent staff are available to cover the duration of infusion. Due to low numbers of epidural use, it cannot be assumed that nurses competent at managing an epidural will be on duty.

If the ward cannot provide competent cover then the anaesthetist must arrange for an alternative ward, able to supply the nursing support required, or utilise an alternative analgesic regimen. For all thoracic epidurals, consideration of a Critical Care admission must be given.

The on-call anaesthetic team MUST be given hand over from the anaesthetist for the first post-operative night. This includes Bleep 504 and the General On-Call Consultant.

General Care

Pre-operative education is essential: patients must be introduced to the concepts of epidural analgesia by the anaesthetist. Nursing staff both on the ward and in the Recovery area will reinforce this information.

The nurse caring for the patient must be a registered nurse who has completed the training and assessment package, or under the supervision of a registered nurse competent in the care of patients receiving epidural analgesia.

The solution must be set up using a clean technique. The pump giving set must be changed every 72hrs.

Observations

Epidural observations must be monitored as per pain observation charts (Appendix B).

Patients may experience a degree of sensory/motor block, even with infusions containing dilute concentrations of local anaesthetics.

The level of sensory block can be checked using the patient's response to cold, and monitoring of motor function using the Bromage Score. Refer to Appendix for guidance on the management of reduced motor function with epidural analgesia.

If a combined spinal / epidural technique is utilised, the patient can be expected to have a motor block for at least 3 hours from the point of insertion. However, after this, Bromage scores MUST be measured and if motor function is not returning then anaesthetic review must be sought as a matter of priority.

The epidural catheter insertion site must be checked 8 hourly for inflammation and leakage and must be recorded on the nursing care plan. Inform the pain team or on-call anaesthetic team if the insertion site is inflamed / painful or if there is excessive leaking.

Mobilising and Motor Blockade

Patients may moblise but must be under the direct supervision of a member of staff.

Lumber epidurals are more likely to cause lower limb weakness. Any patient with an epidural who reports a Bromage Score of 3-4 must be referred to the acute pain team or anaesthetic team immediately.

IV Access

The patient must have a patent intravenous cannula sited for the duration of the epidural infusion in order to administer emergency fluids or drugs. Care for cannula as per the Standard Operating Procedure for performance of peripheral cannulation:

Performance of peripheral cannulation - Standard Operating Procedure **SOP09/004**

Prescriptions

The epidural prescription will be completed, signed and dated by the anaesthetist.

Antiemetic medication and naloxone should be prescribed. Other opioids must not be administered whilst epidural opioids are being infused unless administered by an experienced anaesthetist. Night sedation may be given on consent of the anaesthetist.

Duration / Removal

Unless tunnelled, an epidural catheter should be removed by the end of the fifth day, unless approval has been sought from the anaesthetist who inserted the catheter (see section on removal of catheter, section 11.18). It is essential to ensure that epidural removal is time appropriately with any anti-coagulant therapy. Again, see section 11.18 for further details.

11.5. Trouble-shooting - Poor analgesia

Do not turn down the infusion rate within the first 24 hours, unless the patient is experiencing side effects from the drug infused.

Check catheter insertion site for excessive leakage and migration of catheter, i.e. is the catheter still in the epidural space?

Encourage the patient to utilise the PCEA (Patient Controlled Epidural Analgesia) function and administer and due non-opioid analgesics.

If the pain is severe or not responding to measures, call the Pain Team or on-call anaesthetist.

If a patient requires a bolus dose of local anaesthetic (administered by anaesthetic staff or Pain Team), the blood pressure must be recorded every five minutes for twenty minutes. Sympathetic blockade may lead to hypotension due to vasodilation. Treatment of hypotension is usually by the intravenous infusion of an IV fluid bolus (crystalloid) or by the intravenous injection of ephedrine (doctors only).

11.6. Sedation & Respiratory Depression (Local Anaesthetic with Opioid)

If the respiratory rate is 8-10 breaths per minute and the patient is sedated but easily roused, turn the infusion rate down or stop infusion and give oxygen via facemask at 4 l/min. If the patient is also in pain, consider administration of non-opioid analgesia.

If the respiratory rate drops below 8 breaths per minute or the patient is not easily roused, stop the infusion and manage for opioid toxicity. Also consider potential for local anaesthetic toxicity.

11.7. Respiratory depression (Local Anaesthetic Only)

This rare complication is caused by high local anaesthetic blockade, which affects respiratory nerve fibres. The level of sensory block should be checked and recorded by the Acute Pain Team or on-call anaesthetist. If the patient complains of numbness above the level of the fifth rib, turn off the infusion.

11.8. Nausea and vomiting (Local Anaesthetic with Opioid)

Administer prescribed antiemetic therapy as outlined in the Post Operative Nausea and Vomiting (PONV) Guidelines.

Review antiemetic therapy if nausea / vomiting continue and consider changing or adding in a different antiemetic. Contact Acute Pain Team if problem persists. Always consider a surgical cause, such as bowel obstruction and assess accordingly.

11.9. Hypotension

Local anaesthetic bolus administration will result in hypotension in a patient who is hypovolaemic. Due to vasodilatation it can also occur in well filled patients.

A fluid bolus should be administered, but to a maximum of 1000 ml before vaso-constrictor therapy should be considered. On-going vasoconstrictor therapy requires a critical care assessment and likely admission. Thoracic epidurals have a high risk of hypotensive complications and should always be discussed with Critical Care prior to insertion.

Remember to investigate other causes of hypotension – hypovolaemia and bleeding. Hypotension can be the cause of nausea and vomiting.

Do not stop the epidural infusion without discussing with the acute pain or anaesthetic team.

11.10. Urinary retention

It is advised that all patients are catheterised.

11.11. Pruritis (Local Anaesthetic with Opioid)

Administer prescribed antihistamines or calamine lotion. Consider reducing or removing the opioid from the epidural infusion rate. Administering small increments of naloxone (50-100mcg) IV can often reduce itching without reversing analgesia. Contact the Acute Pain Team if problem persists.

11.12. Local Anaesthetic Toxicity

See Section 13.6

11.13. Dural Tap Headache

Typically, the headache is worse on standing/sitting therefore lie the patient flat or semi-recumbent. Keep them well hydrated and give paracetamol unless contraindicated. Inform the Pain Team as a ‘blood patch’ procedure may be indicated.

11.14. Epidural Abscess

Check epidural site 4 hourly for signs of infection. Obtain advice urgently from the Acute Pain Team or on-call anaesthetist if you are concerned about the following:

- Back pain
- Increasing neurological deficit
- Infected epidural site or pyrexia

11.15. Epidural Haematoma

Obtain advice urgently from the Acute Pain Team or on-call anaesthetist if you are concerned about the following:

- Back pain
- Increasing neurological deficit.

11.16. Management of Motor Weakness with Epidural Analgesia

Refer to Appendix J.

In the event that there is a suspected epidural haematoma, abscess or potential neurological damage as a result of an epidural, it is the responsibility of the anaesthetist who inserted the epidural to review the patient. If the anaesthetist is unavailable then the on call anaesthetic team take on this responsibility.

11.17. Disconnected Epidural Catheter

Wrap the end of the epidural catheter in sterile gauze. Inform the Acute Pain Team or on-call anaesthetist. Do not reconnect the catheter to the filter (stop the infusion). The epidural is likely to need removal and re-siting but this is an anaesthetic decision and should consider timing of any anti-coagulant medication.

11.18. Removal of Epidural Catheter

Anticoagulation

Anti-Coagulant	Minimal Interval Following Dose Before Removing / Inserting Epidural	Minimal Interval Before Administering Drug After Removing Epidural
Enoxaparin (Prophylactic dose)	12 Hours	4 Hours
Enoxaparin (Treatment dose)	24 Hours	4 Hours
Heparin Infusion	4 Hours after infusion stopped No infusion with epidural	4 Hours
Warfarin	INR 1.4 or less No warfarin with epidural	Following day
Dabigatran	5 days	6 Hours
Rivaroxaban	3 days	6 Hours
Clopidogrel	5-7 Days	Following day

Guidelines for Removal

This is an aseptic procedure.

Position the patient in a slightly flexed position in order to open up the intervertebral space.

Remove dressings.

Apply gentle traction to the catheter. The catheter should be removed easily – if there is difficulty, obtain advice from a senior member of staff or the Acute Pain Team.

Ensure that the catheter is intact and the blue tip at the catheter end is identified.

Send the catheter tip to microbiology for MC&S if there are signs of infection at the site, pyrexia or epidural sited for longer than 5 days.

Apply a small clear occlusive dressing to the site.

Remove this dressing after 24 hours.

12. Guidelines for Intrathecal Analgesia

Spinal (Intrathecal) injection, at the lumber level, is suitable for some abdominal surgery e.g. Caesarean section and some lower limb surgery e.g. Total Hip replacement. A spinal injection is made by an anaesthetist under strict aseptic conditions. An Intrathecal injection can contain a combination of drugs: local anaesthetic +/- an opioid (usually diamorphine or fentanyl).

For contraindications, guidelines and trouble shooting section on epidural analgesia.

For specific nursing care please refer to back of Intrathecal prescription chart (Appendix F).

13. Indwelling Nerve Catheters for Infusion / Bolus of Local Anaesthetic (Nerve Blocks)

13.1. Guidance

These guidelines are applicable for indwelling catheters in the regions of:

Femoral / Sciatic nerves

Brachial plexus

Intra-pleural / Para-vertebral spaces

Lumbar / Sacral Plexuses

Abdominal wall – eg TAP or Rectus sheath

Patient selection is by an anaesthetist /surgeon.

All nurses, caring for the patient, must be aware of the side effects of local anaesthetics and potential toxic reactions.

Patient education is essential and should include information regarding the potential side effects of numbness and immobility caused by the action of local anaesthetic.

The patient must have a patent intravenous cannula sited throughout the duration of local anaesthetic catheter placement. This is for the administration of fluids and emergency drugs if required.

Any infusion or bolus dose of local anaesthetic must be prescribed on the patient's prescription chart – an anaesthetist or member of the Pain Team must administer any bolus dose.

Record the patients pulse and blood pressure every 5 minutes for 20 minutes following a bolus dose.

Monitor the patient for signs of local anaesthetic toxicity - see section 13.6

Pain should be scored half an hour following a bolus dose. Otherwise score pain hourly for the first four hours following insertion of the catheter and four hourly thereafter, if stable.

Additional analgesia can be administered including opioids. However, caution and extra monitoring must ensue if the patient has had significant doses of opioid whilst awaiting a regional block / bolus as the subsequent rapid reduction in pain may precipitate opioid toxicity.

If the patient complains of severe pain, contact the Pain Team or on-call anaesthetist. Consider surgical causes of pain such as compartment syndrome.

Observe the insertion site of the catheter for any leakage or signs of inflammation – twice per shift.

13.2. Continuous Nerve Infusion Devices

The continuous nerve infusion device is usually an elastomeric pain relief system. It administers a continuous, regulated flow of local anaesthetics around specifically targeted nerve/nerve plexus for preoperative, perioperative and postoperative pain management and lasts for 48 hours.

The device is set up and filled with local anaesthetic by an anaesthetist but can be connected to the nerve catheter by a member of the Pain Team, anaesthetic team or recovery practitioners.

The device must be clearly labelled with the patient's name, ward, drug and drug concentration. The infusion must be prescribed on the appropriate prescription (see Appendix).

Once the Device is empty the pain relief system and nerve catheter can be removed. Patients are not to be discharged home with a continuous nerve infusion device or nerve catheter. If on-going infusion is required, the Pain Team or on-call anaesthetic team may initiate a further elastomeric device.

If the nerve infusion is anticipated to last more than 48 hours, a Body Guard pump may be used instead using bupivacaine 0.167% (500ml) at 5 ml per hour. The bag must be changed every 72 hours.

13.3. Continuous Wound Infiltration

The continuous wound infiltration device is an elastomeric pain relief system that is intended to provide continuous delivery of local anaesthetic to or around surgical sites for preoperative, perioperative and postoperative pain management. The device is connected to a catheter that is placed under the wound during surgery. The catheter ensures even distribution of local anaesthetic into the wound.

The device is set up and filled with local anaesthetic by an anaesthetist and connected to a catheter by the acute pain team, anaesthetic team or recovery nurses.

The device must be clearly labelled with the patient's name, ward, drug and drug concentration. The infusion must be prescribed on the appropriate prescription (see Appendix G).

Once the device is empty the pain relief system and wound catheter can be removed. The wound catheters may be sutured at site of insertion. Patients are not to be discharged home with a continuous would infiltration device in place.

13.4. Nursing Management

Nursing management of elastomeric local anaesthetic wound infusions is the same as the management of the nerve catheters. In addition, the infusion devices must be checked 4 hourly to ensure it is infusing and clearly documented in the patient's medical notes, along with 4 hourly pain assessment.

13.5. Catheter Removal

This is an aseptic procedure. Catheters can be removed by nursing staff:

Discontinue infusion.

Some catheters are held in place by a suture which should be removed. Apply gentle traction to the catheter. If the catheter proves difficult to remove, inform the Acute Pain Team or the anaesthetist on-call.

Ensure that the catheter is complete with the blue tip identified. (Send catheter tip for MC&S if there are signs of infection).

Cover catheter site with a small occlusive dressing for 24 hours.

13.6. Local Anaesthetic (LA) Toxicity

This occurs with excessive blood circulating levels of local anaesthetic.

Causes of LA toxicity

Inadvertent IV injection

Catheter migration into capillary or vein

Overdose of local anaesthetic

Rapid absorption into the blood stream

Symptoms of Increasing Toxicity

Tingling / Numbness of tongue and lips / Tinnitus

Restlessness / Confusion / Light-headedness

Visual disturbances / Heaviness of limbs / Muscular twitching

Unconsciousness / Generalised convulsions

Cardiovascular collapse / Arrhythmias

Respiratory arrest

Cardiac Arrest - Ventricular Fibrillation (VF) or Asystole

Management of Local Anaesthetic Toxicity

Mild symptoms require no treatment. Discontinue the use of the local anaesthetic, inform the anaesthetist and monitor the patient for any further signs of toxicity.

In a severe reaction (extremely rare and usually due to direct injection of local anaesthetic into a vein) it is important to avoid hypoxia, maintain the circulation and treat any convulsions which occur.

Perform standard resuscitative tasks using an ABC (Airway, Breathing, Circulation) approach. Call for immediate medical and anaesthetic assistance. Administer oxygen and monitor oxygen saturation.

Be prepared to initiate CPR and advanced life-support measures in the event of cardiac arrest. Good quality life support is the key to a successful outcome.

Intralipid 20% emulsion can reverse LA toxicity. It is kept in theatre suites and obstetrics. Intralipid should be administered in cases of significant LA toxicity and always in event of severe cardio-respiratory compromise where LA toxicity is a likely cause.

Intralipid is administered in accordance with AAGBI guidelines (laminates of which should be kept with the drug), but must never interfere with administering high quality CPR: http://www.aagbi.org/sites/default/files/la_toxicity_notes_2010_0.pdf

14. Low Dose Ketamine Infusions

14.1. Background

Ketamine acts as an N-Methyl-D-Aspartate (NMDA) antagonist causing a dissociative anaesthesia at intravenous doses of 1-2 mg / kg (50-200mg). In lower doses of 0.5-1 mg / kg (25-100mg) it is sedative with analgesic properties.

The analgesic properties persist at low doses of 0.1-0.2 mg / kg (5-10mg). If pain persists, intravenous infusions of 0.1-0.2 mg / kg / hour (5-10 mg / hour) for up to 48 hours, can be very effective.

Its use in general anaesthesia has been limited due to hallucinogenic side effects during emergence which can be unpleasant and occur in around 12% of patients. The side effects seem to be dose related and are therefore greatly reduced in low doses. As a result, utilization during the peri-operative period is gaining popularity throughout the UK.

The evidence is supportive of its use, which is becoming increasingly widespread and reviews of scientific evidence have found that at low doses, psychotropic side effects are mild or absent. Ketamine use is beneficial at preventing against 'wind up' and prolonged, severe acute pain increases the risk of acute pain becoming chronic. Ketamine therefore presents an extra tool in managing pain which is poorly responsive to opioids or in patients at risk of developing chronic post-surgical or trauma pain.

Patients whose pain is not well controlled on patient controlled opioid analgesia (PCA) and whose pain extends beyond the scope of regional techniques on offer at that moment may benefit significantly from addition of low dose intravenous ketamine infusions.

Intravenous (IV) Ketamine infusions can safely be used on a general ward in combination with opioid PCA. Ketamine is very safe at the infusion doses of 5-10mg per hour (around 0.1-0.2 mg / kg per hour) with minimal cardiovascular or respiratory effects. In fact, ketamine is associated with increased sympathetic stimulation and can raise blood pressure. The primary risks are those of mild psychotropic phenomena and opioid side effects from the PCA.

14.2. Patient Selection

Only Consultant and SAS grade Anaesthetists may prescribe low dose IV ketamine infusions. Trainee Anaesthetists may prescribe on the advice of the duty Consultant or SAS grade, with the discussion documented in the patient's notes.

Patient selection: Patients whose pain is not well controlled by opioid PCA and in whom regional techniques are not amenable or able to cover the extent of the pain should be considered for low dose ketamine infusion.

Absolute contra-indications for ketamine infusions are:

- Patient refusal
- Allergy to ketamine

Relative contra-indications require the potential for harm from ketamine to be weighed against the negative physiological impact of severe pain and include:

- Previous adverse response to ketamine
- Severe hypertension
- Heart disease where sympathetic stimulation is to be avoided
- Raised intra-cranial or intra-ocular pressures
- History of psychiatric illness with psychosis.

Referral process: Patients should have an opioid PCA already attached and appropriate loading doses administered before referral to the Anaesthetic Department (Acute Pain Team during week day hours). The Acute Pain Team or duty Anaesthetist can then assess whether a low dose ketamine infusion or a regional anaesthetic technique is most appropriate.

Consent: Ketamine for analgesia is an 'off label' use of the drug and is therefore not covered by the manufacturer's license. Doctors prescribing ketamine for analgesia therefore assume responsibility for its use in tandem with the patient. Patients must be involved in the decision process and be made aware of the risk of psychotropic side effects. These are usually in the form of feeling tense, light headed or vivid dreams which may be unpleasant. It is rare to develop frank hallucinations at these doses and the infusion can be stopped at any time if psychotropic effects are developing. This protocol only covers adult patients competent to give their own consent.

14.3. Infusion Dose

The dose for use of IV ketamine infusions outside of Critical Care is 5-10 mg per hour:

Drug	Dose	Total volume using NaCl 0.9%	Concentration (mg / ml)	Rate (ml / hr)
Ketamine Hydrochloride	200 mg	40 ml	5 mg / ml	1-2 ml / hour

There are 2 preparations of Ketamine available in NDDH, 10 mg / ml and 50 mg / ml. The preferred concentration to use is 10 ml / ml:

- One vial of 20 ml Ketamine at 10 mg / ml (total 200 mg) is to be diluted with a further 20 ml of saline 0.9% to a total of 40 ml in a 50 ml syringe, giving a concentration of 5 mg / ml.
- If 10mg / ml ketamine is not available, use 4 ml ketamine at 50 mg / ml (total 200 mg) diluted with a further 36 ml of saline 0.9% to a total of 40 ml in a 50 ml syringe, giving concentration of 5 mg / ml.

The infusion is to be run through a dedicated line at 1-2 ml per hour using the ketamine protocol on a PCA pump. Patients in Critical Care may run the infusion at higher doses if required.

Both the ketamine infusion and PCA opioid, along with a fluid infusion are to be attached via the same cannula using the 3-Way Multiple Infusion set with anti-syphon and anti-reflux valves (also known as the TIVA double set). Each line must be labeled.

Bolus doses: A bolus dose of 5 mg (1 ml) IV ketamine may be administered through the pump. This function is limited to members of the Acute Pain Team or Anaesthetists. A maximum of 2 boluses may be administered in any one hour period with at least 10 minutes between each bolus.

Prescribing: The infusion must be prescribed on the dedicated ketamine infusion chart which is then attached to the drug chart. Anti-emetics, oxygen and naloxone must also be prescribed.

14.4. Monitoring

Full EWS observations must be performed in accordance with the PCA guidelines with the addition of assessment for psychotropic side effects. Patients may require supplementary oxygen for the duration of the infusion.

14.5. Managing Adverse Events

In the event of unpleasant psychotropic side effects, the ketamine infusion is to be stopped and the patient reassured. If the patient is particularly anxious then administration of 1mg intravenous midazolam may be beneficial. Side effects can be expected to subside within 1 hour of cessation of infusion.

Respiratory depression is most likely due to opioid toxicity as ketamine generally preserves respiratory reflexes. Therefore, respiratory depression is managed by stopping the ketamine infusion and managing as per the Acute Pain Management Guidelines for opioid side effects.

Sedation is most likely due to opioid toxicity. Therefore, sedation is managed by stopping the ketamine infusion and managing as per the Acute Pain Management Guidelines for opioid side effects.

15. Entonox

15.1. Introduction

Entonox is a gas containing 50% nitrous oxide and 50% oxygen. Its name is derived from the chemical formulae of the two gases. Entonox has been in use since 1965. The cylinder is identifiable with a blue body and blue and white shoulders. Entonox is a potent analgesic which depends on self-administration and the co-operation of the patient for its successful use. It is quick acting and has an equally rapid offset when administration ceases. It is an ideal agent for short term use and for pain of short duration.

The method of self-administration safeguards the user from overdose from nitrous oxide - when the patient becomes drowsy the mask/mouthpiece falls from the face, thus preventing inhalation.

15.2. Indications for use

Examples include:

Changing dressings and removing drains

Suture / clip removal

Catheterisation

Applying traction to fractures

Application of plaster of paris

Physiotherapy

Pain during childbirth

Suturing wounds

15.3. Contraindications for Entonox Use

Head injuries

Pneumothorax - or at risk of peumothorax. e.g. Initial care of rib fractures

Decompression sickness (the 'bends')/diving accident

Recent underwater dive

Severe bullous emphysema

Bowel obstruction

Intoxication

Maxillofacial injuries

Middle ear surgery

First trimester of pregnancy

15.4. Guidelines for use

Only staff who have undergone Entonox training and who have been assessed as competent may assist the patient in the use of Entonox.

Assess the appropriateness for administration of Entonox by completing the Entonox checklist

Ensure that the Entonox has been prescribed by a doctor.

Check that the equipment is ready to use i.e. cylinder and demand apparatus are available, equipment is clean and in good working order there is sufficient gas to complete the procedure.

Explain the administration of Entonox to the patient - allow them to familiarise themselves with the equipment.

Do not start the procedure until the patient is receiving the full effects of the Entonox - allow them to breathe the gas for at least one minute.

Continually assess the patient during the procedure i.e. pain and sedation. Ensure that the patient continues to obtain analgesia from the use of the Entonox.

When the procedure is completed, close the cylinder valve and exhaust the residual pressure in the system by depressing the purge diaphragm.

The use of a disposable 'single patient use' mouthpiece is recommended. If a patient is using Entonox for more than one day, the antibacterial filter and mouthpiece needs to be changed every 24 hours in accordance with manufactures recommendations.

Document the procedure in the nursing notes.

N.B. The Entonox cylinder must not be left unattended at the patient's bedside. The cylinder must be stored, at room temperature, in an area within the ward or department which is restricted to trust personnel access only.

Patient Instructions:

When using a mouth piece, the patient must breathe through the mouth and not the nose.

Encourage the patient to breathe normally and regularly - do not hyperventilate.

Explain to the patient that they may feel sleepy and/or experience feelings of euphoria, pins and needles or disassociation ('out of body', 'spaced out')

Explain that Entonox is self-administered

15.5. Entonox Special Warnings and Precautions for Use

Entonox should not be used for more than a total of 24 hours, or more frequently than every 4 days, without close clinical supervision and haematological monitoring. Specialist advice should be sought from a haematologist is such cases.

Haematological assessment should include an assessment for megalobastic change in red cells and hypersegmentation of neutrophils. Neurological toxicity can occur without anaemia or macrocytosis and with B12 levels in normal ranges.

In patients taking other centrally acting depressants such as opioids and / or benzodiazepines, concomitant administration of Entonox may result in increased sedation and consequently have effects on respiration, circulation and protective reflexes. If Entonox is to be used in such cases, this should take place under the supervision of appropriately trained staff.

(See BOC, Medical Gases Data Sheet)

15.6. Education and Training

Education and training will be provided by the Acute Pain Team in formal study days and informal training on the ward. Epidural, PCA and Entonox training can also be accessed via the e-learning portal. Competencies will be assessed and written confirmation issued.

16. Monitoring Compliance with and the Effectiveness of the Guideline

16.1. Standards / Key Performance Indicators

- Pain assessment being completed at the required frequency for each patient
- Appropriate analgesia being administered within 30 minutes from assessment

16.2. Process for Implementation

The author consulted with the following stakeholders:

- Consultant Anaesthetists
- Acute Pain Team
- Paediatric Nurses and Doctors
- Pharmacy

- Senior Surgical Nurses
- Drugs and Therapeutics Group

Consultation took the form of a request for comments, feedback via email and face-to-face meetings. Hard copies were available on request.

Approval of the guidelines was given by the Drugs and Therapeutics Group.

The guidelines will be reviewed every three years. The Corporate Governance Manager will send the author a reminder 4 months before the due review date. The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Lead Consultant Anaesthetist for Acute Pain Management. Any significant revisions must be approved by the Drugs and Therapeutics Committee. Because prescription forms are included as appendices, the approval of the Patient Documentation Group will also be sought.

All versions of these guidelines will be archived in electronic format within the Acute Pain Team policy archive by the CNS Acute Pain Management. Revisions to the final document will be recorded on the Document Control Report. To obtain a copy of the archived guidelines, contact should be made with the Acute Pain Team.

16.3. Monitoring and Compliance

Monitoring of implementation, effectiveness and compliance with these guidelines will be subject to audit by the Acute Pain Team. Where non-compliance is found, a member of the Acute Pain Team will provide support and advice to improve practice.

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17.1. General References

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- Role of Ketamine in Acute Postoperative Pain Management: A Narrative Review. Brian M. Radvansky, Khushbu Shah, Anant Parikh, Anthony N. Sifonios, Vanny Le, and Jean D. Eloy: file:///C:/Users/BeardDav/Downloads/749837.pdf

18. Associated Documentation

18.1. Hyperlinks to Relevant Documentation

- Acute Pain Management Policy
- Symptom Management in Palliative care
- British National Formulary
- Performance of peripheral cannulation - Standard Operating Procedure **SOP09/004**
- Post Operative Nausea and Vomiting (PONV) Guideline
- Pain assessment tool

18.2. Appendices

A. Paracetamol Dosing in Children

B. Epidural Infusion Prescription Form

C. Patient Controlled Analgesia Prescription Form (ADULT / >50 Kg)

D. Patient Controlled Analgesia Prescription Form (CHILD <50Kg)

E. Nurse Controlled Analgesia Prescription Form

F. Intrathecal Analgesia Prescription Form

G. Nerve / Wound Catheter Prescription Form

H. Low dose Ketamine infusion Prescription

I. Pain Management Basis Prescribing (Adult) Quick Reference Sheet

J. Management of Leg Weakness with Epidural Analgesia

19. APPENDIX A: PARACETAMOL DOSING IN CHILDREN

Paracetamol use is routine, but with diversity of available administration routes and differing doses which vary by age and weight. The solely age based dose banding used by the BNF for children (BNFc) can lead to over or under dosing in children who are at the top and bottom of the weight range for their age.

As a result, this guideline has adopted a dosing regimen that factors both age and weight. Ensure that the maximum daily dose (maximum dose per 24 hours) is written on the prescription. Prior to administration check all components of the drug chart to identify any previous dose administrations and total dose over the past 24 hours.

Hepatotoxicity is the main concern with paracetamol. To avoid this ensure that the maximum daily doses are not exceeded and a minimum of 4 hours is left between doses. When using 4 hour intervals a maximum of 4 doses per 24 hours may be administered.

Do not administer multiple doses of paracetamol to a child with hepatic impairment without prior discussion with a consultant. Single doses are the same as for healthy children. Malnourished or dehydrated children may accumulate paracetamol and care needs to be taken in such cases.

Dosing for Pyrexia and Discomfort

Pyrexia itself does not require regular paracetamol unless it is associated with discomfort. Always write 'as required' rather than regularly.

Oral and Rectal:

When using less than 6 hour dose intervals, ensure the maximum daily dose (dose in 24 hours) is not exceeded. Write maximum 24 hour dose on drug chart.

Do not use rectal analgesia in neutropenic patients. Round rectal doses *down* to the nearest available suppository dose.

Age	Dose	Frequency	Maximum Daily Dose
Neonate 28-32 Weeks Postmenstrual age	10-15 mg / kg	8-12 Hours As required	30 mg / kg
Neonate >32 Weeks Postmenstrual age	10-15 mg / kg	6-8 Hours As required	60 mg / kg
Infant 1-3 Months	15 mg / kg	4-6 Hours As required	60 mg / kg
Child (>3 months) <50 kg	15 mg / kg	4-6 Hours As required	60 mg / kg
Child >50 kg	1g	4-6 Hours As required	4g

Intravenous:

Prescribe separately to oral prescriptions if <1g. For children <33 kg use the 50 ml vial. Round doses to the nearest 10 mg when prescribing IV.

When using less than 6 hour dose intervals, ensure the maximum daily dose (dose in 24 hours) is not exceeded. Write maximum 24 hour dose on drug chart.

Age	Dose	Frequency	Maximum Daily Dose
Neonate (Preterm >32 Weeks)	7.5 mg / kg	8 Hours	25 mg / kg
Neonate (full term) – Infant <10 kg	7.5-10 mg / kg	6-8 Hours	30 mg / kg
Child 10 - 50 kg (lean weight if obsese)	15 mg / kg	4-6 Hours	60 mg / kg
Child >50 kg (lean weight if obese)	1g	4-6 Hours	4g

Doses for Pain / Peri-Operative Analgesia

Oral:

When using less than 6 hour dose intervals, ensure the maximum daily dose (dose in 24 hours) is not exceeded. Write maximum 24 hour dose on drug chart.

Age	Loading Dose	Maintenance Dose	Maximum Daily Dose
<3 Months	20 mg / kg	20 mg / kg, 8 Hourly or 15 mg / kg, 6 Hourly	60 mg / kg
Child (>3 months) <50 kg	20 mg / kg (max 1g)	15-20 mg / kg, 4-6 Hourly	75 mg / kg or 4g (whichever is lower)
Child >50 kg	1g	1g, 4-6 Hourly	4g

Intravenous:

Prescribe separately to oral prescriptions if <1g. For children <33 kg use the 50 ml vial. Round doses to the nearest 10 mg when prescribing IV.

When using less than 6 hour dose intervals, ensure the maximum daily dose (dose in 24 hours) is not exceeded. Write maximum 24 hour dose on drug chart

Age	Dose	Frequency	Maximum Daily Dose
Neonate Preterm >32 Weeks	7.5 mg / kg	8 Hours	25 mg / kg
Neonate (full term) – 1 Month	10 mg / kg	4-6 Hours	30 mg / kg
Child (> 1 Month) but <50 kg (lean weight if obese)	15 mg / kg	4-6 Hours	60 mg / kg
Child >50 kg (lean weight if obese)	1g	4-6 Hours	4g

Rectal:

When using less than 6 hour dose intervals, ensure the maximum daily dose (dose in 24 hours) is not exceeded. Write maximum 24 hour dose on drug chart.

Do not use rectal analgesia in neutropenic patients.

Age	Loading Dose	Maintenance Dose	Maximum Daily Dose
Neonate >32 Weeks Up to 1 Month	30 mg / kg	20 mg kg 8 Hourly	60 mg /kg
1-3 Months	30 mg / kg	15 mg / kg 6 Hourly	75 mg /kg
Child (>3 Months) but <50 kg (lean weight if obese)	40 mg / kg	15 mg / kg 4-6 Hourly	75 mg / kg Max 4g
Child >50 kg (lean weight if obese)	1g	1g 4-6 Hourly	Max 4g

Discharge Medication

Any child being discharged home should be advised to use the dose guidance on the bottle.

20. APPENDIX B: Epidural Infusion Prescription

Patient Identification Label

EPIDURAL INFUSION PRESCRIPTION

Secure to patient's drug chart
May use additional opioid PCA if Bupivacaine 0.167% only prescription

Do not administer Enoxaparin for 4 hours after epidural catheter insertion or removal

Do not insert or remove epidural catheter <12 hours after administering Enoxaparin

Do not start Warfarin, Heparin, Clopidogrel or Enoxaparin >40mg whilst epidural in situ

Date	Ward	Allergies	Weight (if <50 Kg)

LEVEL OF INSERTION: **DEPTH OF CATHETER TO SKIN:** **cm**

Clinician boluses may be administered by Anaesthetic, Pain Team and Recovery staff

Higher concentration boluses by Anaesthetists only

LOCAL ANAESTHETIC TOXICITY:

Mild / Moderate toxicity:

Tingling around mouth / Metal taste / Ringing ears / Dizziness / Blurred vision / Muscle twitches

CLAMP the local anaesthetic infusion pump and contact the Acute Pain Team / Anaesthetics

Severe toxicity:

Sudden loss of consciousness / seizures / Cardiac arrest

*Provide supportive care and **Advanced Life Support**. Consider Intralipid Emulsion 20% **in addition** to CPR*

DATE	TIME	PRESCRIPTION	Sign / Print /GMC (Anaesthetists only)
		BUPIVACAINE 0.167% 500 ml via Body Guard™ Pump at 6-12 ml / hr	
		BUPIVACAINE 0.1% + Fentanyl 2 mcg / ml 500 ml via Body Guard™ Pump at 6-12 ml / hr	
		PATIENT CONTROLLED OPTION (PCEA) 5ml of selected bupivacaine solution, 20 minute lock out	
		CLINICIAN BOLUS VIA PUMP: 5-10 ml (record bolus delivery on rear of sheet)	

Patient Identification Label

EPIDURAL INFUSION RECORD:

Date					
Time					
Volume (ml)					
Rate (ml)					
Bupivacaine 0.167%	Yes / No	Yes / No	Yes / No	Yes / No	Yes / No
Bupivacaine 0.1% + Fentanyl 2 µg / ml	Yes / No	Yes / No	Yes / No	Yes / No	Yes / No
Prepared by / Checked by					
Connected by					
Infusion wasted (ml)					

CLINICIAN BOLUSES ADMINISTERED:

Date								
Time								
Volume								
Sign								

EPIDURAL OBSERVATIONS:

Initial set up / following bolus: Full EWS 15 minutes after connection for 1 hour
Next 2 Hours: Full EWS every 30 minutes
On-Going: Full EWS at start of each shift and 4 hourly

Maintain IV access for duration of infusion.
Check epidural catheter 4 hourly for leakage, infection and displacement

Date												
Time												
Rate												
Bolus: Attempted / Delivered												
Total Infused												
Programme check												
Site check												
Bromage: Left / Right												
Pain Score 0-10												
Sign												

CONTACT FOR ADVICE

Acute Pain Management Guidelines (BOB Intranet / Policies / A-Z)
Ward doctors
Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
First on-call Anaesthetist: **Bleep 504**

21. APPENDIX C: Patient Controlled Analgesia Prescription Form (ADULT / >50 Kg)

Patient Identification Label	PATIENT CONTROLLED ANALGESIA (PCA) PRESCRIPTION ADULT / CHILD >50Kg

Secure to patient's drug chart + Ensure antiemetic and O_2 prescribed
Further opioids only on advice of Anaesthetics or Acute Pain Team

Date:	Ward:	Allergies:	Weight (Kg):

DATE	TIME	PRESCRIPTION	Continuous	Sign / Print / GMC (Anaesthetists only)
		MORPHINE SULPHATE (1mg / ml) 50mg made to 50ml with 0.9% NaCl ..... mg (0.5-2 mg) per 5 minutes*	☐ .... ml / hr	
		FENTANYL (10 micrograms / ml) 500micrograms made to 50ml with 0.9% NaCl ..... mcg (5-20 mcg) per 5 minutes*	☐ .... ml / hr	
		OXYCODONE (1mg / 1ml) 50mg made to 50ml with 0.9% NaCl ..... mg (0.5-2 mg) per 5 minutes*	☐ .... ml / hr	
		Pump Administered Bolus up to ..... ml (..... micrograms / mg) in 2-5 ml increments		
		Naloxone 200 micrograms IV As Required to reverse toxicity		

*0.5-1 ml = 1 minute bolus + 4 minute lock, 2 ml = 2 minutes bolus + 3 minute lock.

PUMP BOLUS DOSES ADMINISTERED

Pump administered boluses may be delivered by Anaesthetists, Pain Team and Recovery staff

Total Dose								
Date								
Time								
Signature								

MANAGING COMPLICATIONS

Respiratory rate <8 per minute or Excessive Sedation:

1. STOP infusion. Assist breathing as required (Open airway, O_2, Bag / Valve / Mask ventilate, call 2222)
2. Administer Naloxone 200micrograms Slow IV Stat
   (400 micrograms Naloxone in 1 ml+ 9 ml normal saline = 40 mcg/ml, 5 mls = 200mcg)
   Repeat every 3 – 5 mins as necessary until patient alert

Patient Identification Label

DRUG INFUSION RECORD

Date	Time	Drug / Dose / Volume	Prepared by:	Checked by:	Total Infusion wasted (ml) & sign.

PCA OBSERVATIONS

Initial set up / bolus: Full EWS 15 minutes after loading dose
Followed by: Full EWS every 30 minutes for two hours and 1 hourly for 4 hours
On Going: Full EWS at start of each shift and 4 hourly (More frequent if indicated)

Complete PCA specific data with each set of EWS observations (max 1 hourly)

Date	Time	Good demands	Attempts	Total volume Infused	Programme check	VIP score	Pain Score 0-10	Sign

CONTACT FOR ADVICE

Pain Management Guidelines (BOB Intranet – under Acute Pain Management)
Ward doctors
Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
First on-call Anaesthetist: **Bleep 504**

22. APPENDIX D. Patient Controlled Analgesia Prescription Form (CHILD <50Kg)

Patient Identification Label

PATIENT CONTROLLED ANALGESIA (PCA) PRESCRIPTION CHILD <50Kg

Secure to patient's drug chart + Ensure antiemetic and O_2 prescribed
Further opioids only on advice of Anaesthetics or Acute Pain Team

Date:	Ward:	Allergies:	Weight (Kg):
	Caroline Thorpe / ICU		

DATE	TIME	PRESCRIPTION	Continuous	Sign / Print / GMC (Anaesthetists only)
		MORPHINE SULPHATE (20 mcg / Kg /ml) 1 mg / Kg of Morphine Sulphate Made to 50 ml with 0.9% NaCl **..... ml (0.5-1 ml) per 5 minutes***	**0.2 ml / hr** **Sign:**	
		OXYCODONE (20 mcg / Kg / ml) 1 mg / Kg of Oxycodone Made to 50 ml with NaCl 0.9% solution **..... ml (0.5-1 ml) per 5 minutes***	**0.2 ml / hr** **Sign:**	
		Pump Administered Bolus up to ..... ml **(..... micrograms / mg) in 2-5 ml increments**		
		Naloxone .... mcg IV (10 mcg / Kg) **As Required to reverse toxicity**		

*** 1 minute bolus + 4 minute lock**

PUMP BOLUS DOSES ADMINISTERED

Pump administered boluses may be delivered by Anaesthetists, Pain Team and Recovery staff

Total Dose								
Date								
Time								
Signature								

MANAGING COMPLICATIONS

Respiratory Depression:

Respiratory rate <20 / min (1-5 years), <16 / min (5-12 years), <10 / min (12+ years)

STOP infusion. Assist breathing as required
(Open airway, O_2, Bag / Valve / Mask ventilate, call 2222)
Administer Naloxone 10 mcg / Kg Slow IV Stat
Repeat every 3 – 5 minutes, as necessary, until patient alert

Patient Identification Label

DRUG INFUSION RECORD

Date	Time	Drug / Dose / Volume	Prepared by:	Checked by:	Total Infusion wasted (ml) & sign.

PCA OBSERVATIONS

Initial set up / bolus: Full EWS 15 minutes after loading dose
Followed by: Full EWS every 30 minutes for two hours and 1 hourly for 4 hours
On Going: Full EWS at start of each shift and 4 hourly (More frequent if indicated)

Complete PCA specific data with each set of EWS observations (max 1 hourly)

Date	Time	Good demands	Attempts	Total volume Infused	Programme check	VIP score	Pain Score 0-10	Sign

CONTACT FOR ADVICE

Pain Management Guidelines (BOB Intranet – under Acute Pain Management)
Ward doctors
Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
First on-call Anaesthetist: **Bleep 504**

23. Appendix E. Nurse Controlled Analgesia Prescription Form

Patient Identification Label

NURSE CONTROLLED ANALGESIA (NCA) PRESCRIPTION

Secure to patient's drug chart + Ensure antiemetic and O_2 prescribed
Further opioids only on advice of Anaesthetics or Acute Pain Team

Date:	Ward:	Allergies:	Weight (Kg):

DATE	TIME	PRESCRIPTION	Continuous	Sign / Print / GMC (Anaesthetists only)
		MORPHINE SULPHATE 1 mg / Kg Made to 50 ml with 0.9% NaCl (If >50 Kg use 50 mg in 50 ml) **1 ml per 20 minutes***	**1 ml / hr** **Sign:**	
		OXYCODONE 1 mg / Kg Made to 50 ml with 0.9% NaCl (If >50 Kg use 50 mg in 50 ml) **1 ml per 20 minutes***	**1 ml / hr** **Sign:**	
		Pump Administered Bolus up to 5 ml		
		Naloxone .... mcg IV (10 mcg / Kg) **As Required to reverse toxicity**		

***1 minute bolus + 19 minutes lock**

PUMP BOLUS DOSES ADMINISTERED

Pump administered boluses may be delivered by Anaesthetists, Pain Team and Recovery staff

Total Dose								
Date								
Time								
Signature								

MANAGING COMPLICATIONS

Respiratory Depression:

Respiratory rate / min: <20 (1-5 years), <16 (5-12 years), <10 (12+ years), <8 (Adult)

STOP infusion. Assist breathing as required
(Open airway, O_2, Bag / Valve / Mask ventilate, call 2222)

If Necessary Administer Naloxone 10 mcg / Kg Slow IV Stat
Repeat every 3 – 5 minutes, as necessary, until patient alert

Patient Identification Label

DRUG INFUSION RECORD

Date	Time	Drug / Dose / Volume	Prepared by:	Checked by:	Total Infusion wasted (ml) & sign.

NCA OBSERVATIONS

Initial set up / after bolus dose: Full EWS 15 minutes
Followed by: Full EWS every 30 minutes for two hours and 1 hourly for 4 hours
Subsequently: Full EWS at start of each shift / 4 hourly (More frequent if indicated)

Complete NCA specific data with each set of EWS observations (max 1 hourly

Date	Time	Good demands	Attempts	Total volume Infused	Programme check	VIP score	Pain Score 0-10	Sign

CONTACT FOR ADVICE

Pain Management Guidelines (BOB Intranet – under Acute Pain Management)
Ward doctors
Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
First on-call Anaesthetist: **Bleep 504**

24. APPENDIX F. Intrathecal Analgesia Prescription Form

Patient Identification Label

INTRATHECAL (SPINAL) ANALGESIA ADMIINSTRATION CHART

(KEEP WITH PATIENT DRUG PRESCRIPTION)

Date		Ward	
Procedure		Weight	
Anaesthetist		Allergies	

INTRATHECAL DRUGS ADMINISTERED

	Local Anaesthetic (ml and concentration)	Opioid (drug and dose)
Intra-Thecal Drugs		
Dose / Volume		
Time Given		
Signature		
Name / GMC		

FOLLOW GUIDELINES OVERLEAF IF ADDITIONAL OPIATES REQUIRED

ENSURE OXYGEN, NALOXONE & ANTI-EMETICS ARE PRESCRIBED

Patient Identification Label

DRUGS AND DURATION OF ACTION

- Intra-thecal levobupivacaine induces spinal anaesthesia - weak / numb legs for 2-4 hours.
- Intra-thecal Fentanyl gives around 2-6 hours opioid affect
- Intra-thecal Diamorphine gives around 12-16 hours opioid affect

SIDE EFFECTS OF INTRATHECAL OPIATES

- Sedation / Respiratory depression / Oxygen de-saturation,
- Nausea
- Pruritis
- Urine retention.
- Hypotension (less common than with epidural PCA)

COMPLICATIONS OF SPINAL INJECTION

Severe frontal headache occasionally occurs after an intra-thecal injection (more common in younger age group e.g. post caesarian section).

Severe back pain and leg weakness: This is very rare and may indicate a spinal haematoma or abscess (and needs urgent medical attention).

RESPIRATORY DEPRESSION

If respiratory rate < 8 breaths/minute or patient becomes un-rousable: call for help
If necessary, assist breathing with bag / mask / oxygen and call the cardiac arrest team
Administer Naloxone

OBSERVATIONAL ASSESSMENT

Monitor respiratory rate and SpO_2 continuously for 15 hours after diamorphine.

Record Observations:

- Every 10minutes in Recovery
- Half hourly for 2 hours on return to ward
- Hourly and then 4 hourly as appropriate

Routine observations: blood pressure, pulse, respiratory rate and oxygen saturations

Record pain score on movement and deep breathing

Record and document sedation score

Monitor temperature for signs of infection

Maintain intravenous access

INADEQUATE ANALGESIA

Most patients will require regularly prescribed oral analgesia; if additional analgesia is required consult with duty anaesthetist.

CONTACT FOR ADVICE

Acute Pain Management Guidelines (BOB Intranet / Policies / A-Z)
Ward doctors
Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
First on-call Anaesthetist: **Bleep 504**

25. APPENDIX G. Nerve / Wound Catheter Prescription Form

Patient Identification Label

NERVE / WOUND CATHETER INFUSION PRESCRIPTION CHART

Secure to patient's drug chart
A maximum of 2 Local Anaesthetic infusions may be used simultaneously
Administer prescribed analgesia in addition - may include opioids

Date	Ward	Allergies	Weight (if <50 Kg)

DATE	TIME	POSITION AND SIDE OF CATHETER	PRESCRIPTION	Sign / Print / GMC (Anaesthetists only)
		LEFT / RIGHT / MIDLINE:	BUPIVACAINE 0.167% 250 ml in *Elastomeric balloon pump* 5 ml / hr	
		LEFT / RIGHT / MIDLINE:	BUPIVACAINE 0.167% 250 ml in *Elastomeric balloon pump* 5 ml / hr	
		LEFT / RIGHT / MIDLINE:	BUPIVACAINE 0.167% 500 ml via *Body Guard™ Pump* 5 ml / hr	
		LEFT / RIGHT / MIDLINE:	BUPIVACAINE 0.167% 500 ml via *Body Guard™ Pump* 5 ml / hr	
		LEFT / RIGHT / MIDLINE:	BOLUS: LEVOBUPIVACAINE 0.25% 5-10 ml, As Required, Max 6 Hourly	
		LEFT / RIGHT / MIDLINE:	BOLUS: LEVOBUPIVACAINE 0.25% 5-10 ml, As Required, Max 6 Hourly	

Patient Identification Label

NERVE / WOUND CATHETER OBSERVATIONS

Initial set up: Full EWS 15 minutes after connection for 1 hour
Followed by: Full EWS every 30 minutes for two hours
Subsequently: Full EWS at start of each shift and 4 hourly
Maintain IV access for duration of infusion.
Check catheter 4 hourly for leakage, infection, displacement and residual volume.

BUPIVACAINE 0.167% INFUSION RECORD

Date	Time	Vol (ml)	Rate (ml / hr)	Pump (Elatomeric / Body Guard)	Catheter Position	Connected by	Checked by
					LEFT / RIGHT / MIDLINE		
					LEFT / RIGHT / MIDLINE		
					LEFT / RIGHT / MIDLINE		
					LEFT / RIGHT / MIDLINE		

BOLUS DOSES (Acute Pain / Anaesthetics Only)

Date								
Time								
Site								
Vol (ml)								
Sign Print Registration								

DISCONTINUATION OF CATHETER

Date/Time	Stopped by	Removed by	Reason	Notes

LOCAL ANAESTHETIC TOXICITY

Mild / Moderate toxicity:
Tingling around the mouth / Metallic taste / Ringing in the ears / Dizziness / Blurred vision / Muscle twitches
CLAMP the local anaesthetic infusion pump and contact the Acute Pain Team / Anaesthetics

Severe toxicity:
Sudden loss of consciousness / seizures / Cardiac arrest
*Provide supportive care and **Advanced Life Support**. Consider Intralipid Emulsion 20% **in addition** to CPR.*

CONTACT FOR ADVICE

1. Acute Pain Management Guidelines (BOB Intranet / Policies / A-Z)
2. Ward doctors
3. Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
4. First on-call Anaesthetist: **Bleep 504**

26. APPENDIX H. Low dose Ketamine infusion Prescription

Patient Identification Label

ANALGESIC LOW DOSE IV KETAMINE INFUSION PRESCRIPTION CHART

Secure to patient's drug chart
Document consent process in notes

Date	Ward	Allergies	Weight (if <50 Kg)

DATE	TIME	PRESCRIPTION	Infusion Rate	Sign & Print (Anaesthetists only)
		KETAMINE (5mg / ml) 200 mg made to 40 ml with 0.9% NaCl 1-2 ml (5-10 mg) per hour	.... ml / hr	
		KETAMINE BOLUS 5 mg (1 ml of infusion) Maximum of 2 boluses per hour		

BOLUS DOSES ADMINISTERED

(Only Acute Pain Team or Anaesthetists to Administer a Ketamine Bolus)

Total Dose								
Date								
Time								
Signature								

KETAMINE INFUSION RECORD

Date	Time	Dose (mg) / Volume (ml)	Rate (ml / hr)	Prepared by	Checked by	Total Infusion wasted (ml) & sign

Patient Identification Label

KETAMINE OBSERVATIONS

Ketamine is used in combination with opioid PCA and managed as per PCA with the addition of assessing for hallucinogenic effects

After bolus: Full EWS after 15 minutes (only anaesthetists to administer boluses)

Initial set up: Full EWS every 30 minutes for two hours and 1 hourly for 4 hours

On Going: Full EWS at start of each shift and 4 hourly (More frequent if indicated)

Complete ketamine specific data with each set of EWS observations (max 1 hourly):

Date	Time	Total Volume Infused (ml)	Hallucinogenic effects (nil / mild / unpleasant)	Pain Score 0-10	Sign

MANAGING COMPLICATIONS

Hallucinogenic effects:

Mild – consider reducing rate or continue if patient untroubled

Unpleasant – Reassure. Reduce or stop infusion depending on patient preference.

Consider midazolam 1mg IV if distressed.

Respiratory Rate <8 per minute: Manage as PCA opioid toxicity

Excessive sedation: Stop ketamine. Manage as PCA opioid toxicity

Nausea: Administer anti-emetic

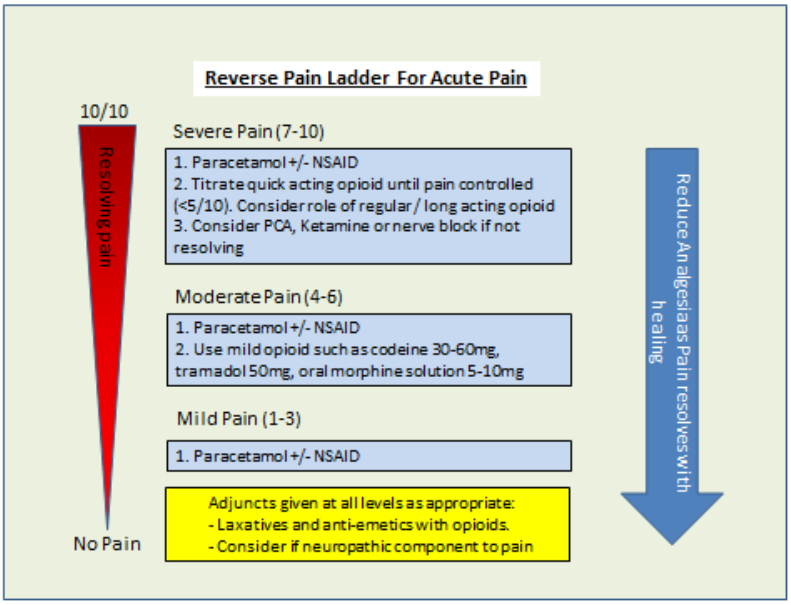
Pruritis: Administer anti-histamine

If necessary contact Acute Pain Team or ward doctors to review

CONTACT FOR ADVICE

1. Acute Pain Management Guidelines (BOB Intranet / Policies / A-Z)
2. Ward doctors
3. Acute Pain Team (08:00 – 16:00 Monday – Friday): **Bleep 279**
4. First on-call Anaesthetist: **Bleep 504**

27. APPENDIX I. PAIN MANAGEMENT BASIC PRESCRIBING (ADULT) QUICK REFERENCE SHEET



Paracetamol

Weight <50 kg, malnutrition, liver disease or dehydration - reduce the dose: 750 mg or 500 mg oral dosing, by weight for IV.

Route	Dose	Frequency	Maximum Daily Dose
Oral	1g	4-6 Hours	4g
IV	1g	4-6 Hours	4g
Rectal	1g	4-6 Hours	4g
IV (<50 kg)	15 mg / kg	4-6 Hours	60 mg / kg

Ibuprofen

Route	Dose Ibuprofen	Frequency
Oral (>50 kg)	400 mg	6-8 hourly
Oral (< 50 kg)	7.5 (5-10) mg / kg	6-9 hourly (max 30 mg / kg / day)

Morphine

Patient Factors	Route	Dose	Frequency
Frail / Elderly / Opioid sensitive	Oral	5-10 mg	2-4 Hours 'As required'
Frail / Elderly / Opioid sensitive	IV / IM / SC	2.5-5 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	Oral	10-20 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	IV / IM / SC	5-10 mg	2-4 Hours 'As required'

Oxycodone

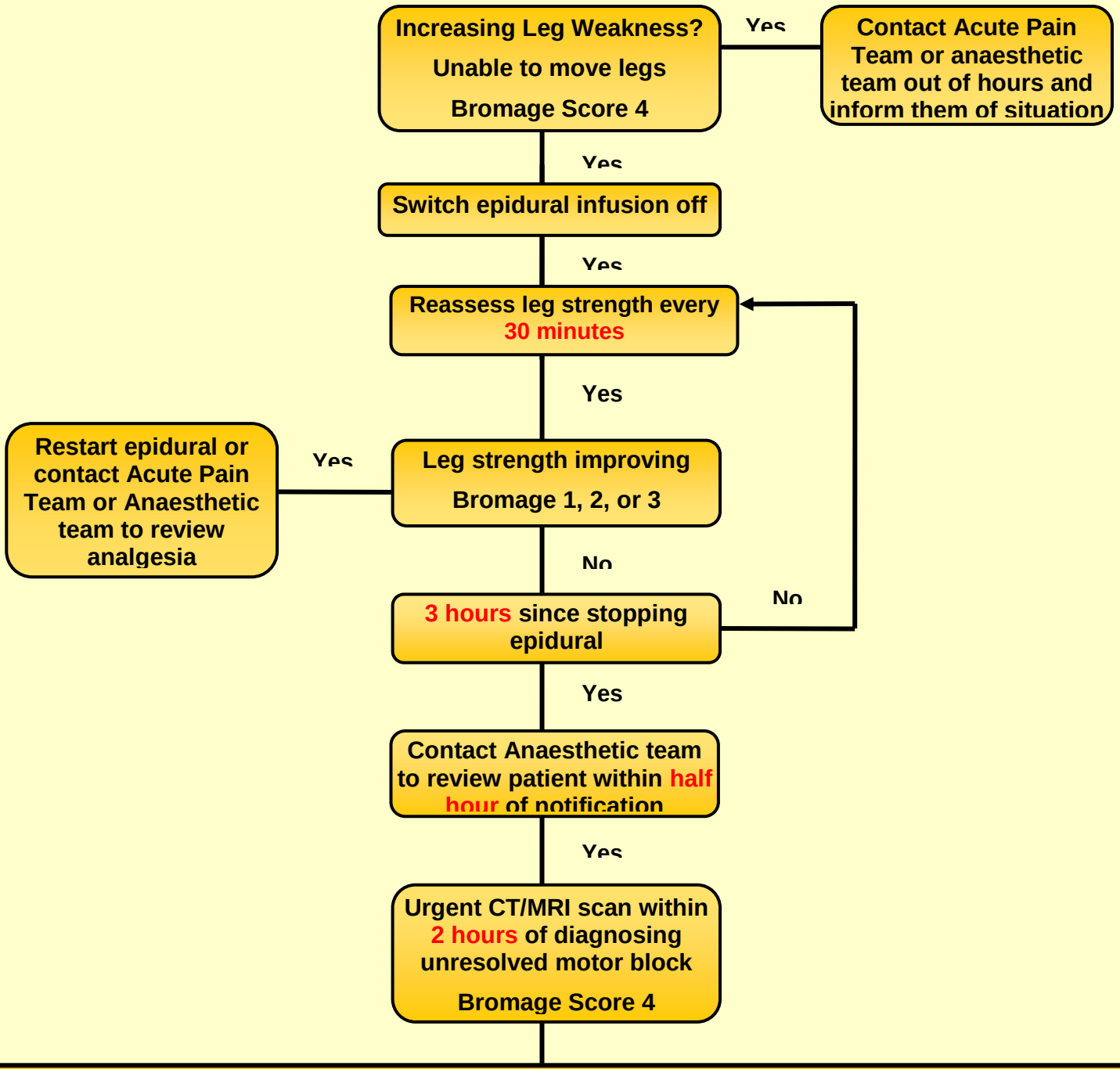
eGFR<30 or morphine intolerant

Patient Factors	Route	Dose	Frequency
Frail / Elderly / Opioid sensitive	Oral	2.5-5 mg	2-4 Hours 'As required'
Frail / Elderly / Opioid sensitive	IV / IM / SC	2.5 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	Oral	5-10 mg	2-4 Hours 'As required'
Robust / Not opioid sensitive	IV / IM / SC	2.5-5 mg	2-4 Hours 'As required'

28. APPENDIX J. Management of Leg Weakness with Epidural Analgesia

This is to be printed and posted as a reminder of basic doses of analgesia.
Refer to full guidelines for further advice.

All patients receiving epidural analgesia must have leg strength assessed and documented. Thoracic epidural analgesia should not cause profound leg weakness. Increasing leg weakness usually means the infusion rate is too high. However it may mean that the patient is developing an **epidural haematoma**. If not diagnosed and treated promptly, this will lead to paraplegia. This algorithm should help in the management and treatment of suspected epidural haematomas.



An epidural haematoma has to be evacuated within 8 hours of the onset of symptoms (when first suspected) for your patient to have the best chance of recovery of neurological function. Be aware that patients with an epidural haematoma may need to be transferred to a specialist unit, hence the urgency for assessment and diagnosis. Within North Devon we have stated a maximum time of 6 hours between onset of symptoms and diagnosis to allow time for transfer. Do not delay.